

1 Introduction to Nanocarriers for Nucleic Acids and Proteins

Juan C. Cruz, Yashwant V. Pathak, and Luis H. Reyes

1.1 INTRODUCTION TO NANOBIOTECHNOLOGY

Nanobiotechnology merges nanotechnology and biology to address complex biomedical challenges. At the nanoscale (1–100 nm), materials exhibit unique properties that enable precise manipulation of biological systems. In drug delivery, nanoparticles are engineered to target specific cells or tissues, enhancing therapeutic efficacy while minimizing side effects. Notably, lipid nanoparticles (LNPs) have been instrumental in delivering mRNA vaccines against COVID-19, revolutionizing vaccine development (Hou et al., 2021). These nanoparticles improve drug solubility and bioavailability, opening avenues for treating previously intractable diseases.

Nanobiotechnology also advances diagnostic techniques. Nanosensors and quantum dots enable high-sensitivity detection of biomarkers at early disease stages, crucial for timely intervention in conditions like cancer and infectious diseases (Khazaei et al., 2023). In regenerative medicine, nanomaterials mimic the extracellular matrix, promoting cell growth and tissue regeneration. These bio-nano scaffolds are essential in developing artificial tissues and organs, potentially alleviating the shortage of donor organs (Fathi-Achachelouei et al., 2019). Nanostructured biomaterials integrate seamlessly with human tissues, advancing organ repair and regeneration.

Recent developments include the exploration of nanorobots for precise drug delivery and surgical procedures, pushing the boundaries of medical innovation (Ye & Sun, 2023). These advancements underscore nanobiotechnology's profound impact on medicine—from precise drug delivery to innovative diagnostics and regenerative solutions. As research progresses, the field continues to bridge nanotechnology and biology, addressing some of today's most pressing healthcare challenges.

These tiny scales confer on materials a sufficiently higher surface area and quantum effects that enable them to engage biological systems with great specificity. Such specificity can become of much consequence when one considers therapeutic agents like nucleic acids—siRNA, mRNA, and plasmid DNA—that are constitutively labile and subject to enzymatic digestion within the body. Enter nanocarriers tasked with safeguarding these fragile molecules just so they reach their targets intact. For instance, LNPs have transformed nucleic acid delivery by encapsulating genetic materials from degradation and thus offering an easier way of entering the cells. That approach has been very successful in the mRNA vaccines against COVID-19, whereby safe and efficient delivery of mRNA was delivered by LNPs (Hou et al., 2021).

Another pathway for nucleic acid delivery involves polymeric nanoparticles made from biodegradable substances, including poly(lactic-co-glycolic acid) (PLGA). By decorating their surfaces with targeting ligands, such nanoparticles are capable of targeting specific cells, hence resulting in enhanced uptake and effectiveness of the therapeutic agents carried therein (Rehman et al., 2022). In the tree-like branched structures of dendrimers lies a flexible platform for binding nucleic acids. Their design thus allows for control over size and functionality with extreme care, hence enhancing delivery precision. This technology also benefits proteins, which are very important but at the same time sensitive therapeutic agents (Mintzer & Grinstaff, 2010).

Nanogels protect the proteins with a hydrophilic network to prevent degradation, thus allowing for controlled release. Among all nanoparticles, gold nanoparticles have emerged as the most biocompatible and easily functionalized ones. They can deliver proteins inside cells directly for increased therapeutic efficiency with minimum side effects (Zhang et al., 2012). Nanoparticles of porous silica are capable of taking up substantial quantities of proteins and releasing them in a stimulus-specific fashion, demonstrating the potential for release upon changes in pH (Mohan et al., 2024). Nanocarrier design is much more than protection and delivery; these are highly specialized mechanisms for overcoming biological barriers. With attachments of molecules like antibodies or peptides on the surface, nanocarriers achieve targeted delivery—secret attack—only the diseased cells are under attack while leaving others healthy. Targeted drug delivery does not only maximize therapeutic effectiveness but also minimizes side effects (Yang et al., 2024).

Besides, nanocarriers are engineered to allow endosomal escape. Upon cellular uptake, they can disrupt the endosomal membranes and release into the cytoplasm before being degraded. Coating usually occurs with substances such as polyethylene glycol (PEG), which extends their circulation time in the body by giving them stealth properties because of reduced recognition and clearance through the immune system (Gisbert-Garzarán et al., 2021). Emerging platforms in nanobiotechnology are opening up new frontiers. Exosomes are natural vesicles secreted from cells that, nowadays, are being developed into biocompatible nanocarriers with inherent targeting properties. They can carry nucleic acids and proteins across all biological barriers and have therefore emerged as a promising path toward various future therapies (Kalluri & LeBleu, 2020).

Similarly, DNA origami nanostructures are an advanced approach in which the DNA molecules are folded into well-defined shapes instead of their normal random structure and can encapsulate therapeutic agents, releasing them in response to specific triggers. These deliveries point to the huge leap in concepts concerning therapeutic delivery. Through the principles of nanobiotechnology, we can design nanocarriers that work in harmonious relay with the biological entity of the body, thus opening new vistas for the treatment of diseases that, until now, were considered beyond the realm of treatment. This stream of continuous research and innovations has brilliant future prospects in medicine and might change the whole treatment and patient care policies.

The development of nanobiotechnology represents a chronicle with landmark, pathbreaking discoveries and inventions forming the foundation that molded the topological view of biomedical applications into their current forms. It all began with the prophetic lecture of Richard Feynman in 1959 entitled “Plenty of Room at the Bottom,” where he mentioned the concept of manipulating matter with regard to atoms (Feynman, n.d.). Thus, the idea sowed the seeds for future excursions into the realm of nanoscale dimensions. Until the discovery of Scanning Tunneling Microscopy (STM) by Gerd Binnig and Heinrich Rohrer in the 1980s, a breakthrough that finally made it possible for scientists to see single atoms, there was no progress toward this dream (Binnig & Rohrer, 1984). This was an important invention in the field of microscopy that finally gave way to the tools that had to be used to manipulate and understand material at the nanoscale level.

As the field matured, the development of Atomic Force Microscopy (AFM) further expanded the capability to study biological molecules in unprecedented detail, thus extending our capabilities to study in great detail a wide range of biological molecules. Advances in materials science and microscopy opened up doors for novel approaches in drug delivery and diagnostics. With the beginning of the 1990s, liposomal drug delivery systems started to be marketed, and in 1995 the Food and Drug Administration (FDA) approved Doxil, a liposomal doxorubicin, as the first nanomedicine on the market (Barenholz, 2012). The success of Doxil showed how nanocarriers can improve the therapeutic index of existing drugs by enhancing delivery to target tissues while minimizing systemic toxicity.

With this momentum, different nanocarrier platforms were developed to include polymeric nanoparticles, dendrimers, and quantum dots that enabled controlled release and improved targeting of therapeutic agents. Farokhzad and Langer (2009) further developed gold nanoparticles and carbon nanotubes that added to the toolkit of biomedical applications and provided unique optical and electronic properties that could be harnessed for imaging and sensing (Farokhzad & Langer, 2009). In recent years, this area began to reach new heights with the appearance of mRNA vaccines using LNPs as vectors (Dreaden et al., 2011). The rapid development and deployment of COVID-19 vaccines by Pfizer-BioNTech and Moderna underlined the transformative potential of nanocarriers in therapeutics (Horejs, 2021). These LNPs protect fragile mRNA molecules and facilitate their uptake by host cells to show their nanobiotechnology contribution to critical participation in modern medicine.

Each of these historical milestones has helped shape the modern landscape in nanocarrier technology for biomedical applications. From Feynman’s early musings to today’s sophisticated nanoengineered systems, the journey has been one of relentless innovation-seeking at the foot of nanotechnology–biology hybrids.

Biological systems fundamentally function on a nanoscale architecture, as proteins, nucleic acids, and cellular organelles possess nanometric dimensions. The intrinsic operation at the nanoscale implies that the nanoparticulation of carriers aligns their dimensions with those of biological molecules, facilitating seamless interactions. By operating at the nanoscale, we can create carriers that can be designed to mimic natural biomolecules, thereby allowing them to interact not only more effectively but also with great specificity.

One of the major advantages is brought about by size compatibility. Nanocarriers, owing to their minuscule dimensions, can cross through complex biological barriers within the body, such as cellular membranes, to intracellular targets where nucleic acids and proteins carry out their functions (Salvati et al., 2011). Their size enables them to take advantage of existing cellular uptake mechanisms, such as endocytosis, thus increasing the rate of cellular internalization and delivery efficiency.

The larger nanocarriers possess a significantly greater surface area relative to their volume. With an increase in the superficial area, there is a higher functionalization with targeting ligands, polymers, or other therapeutic molecules. Thus, the nanocarriers can be engineered so that they identify given cell types and receptors, and bind with or within them, hence enhancing specific targeting and reducing off-target effects (Duncan & Gaspar, 2011).

In other words, with nanoscale carriers, unique physicochemical properties not observable in bulk materials may be expressed. This includes enhanced permeability—through the crossing of biological barriers like the blood–brain barrier (BBB), which is of vital importance for the treatment of central nervous system disorders (Saraiva et al., 2016). More importantly, nanocarriers can protect degradable biological molecules, including nucleic acids, from nucleases and proteases in the bloodstream to ensure that more of the therapeutic agent reaches its target (Peer et al., 2007).

These nanodimensions in size let the carriers take advantage of the enhanced permeability and retention (EPR) effect observed in tumor tissues; tumors usually have leaky vasculature that permits nanocarriers to accumulate with much more ease at tumor sites. This increases the local concentration of the therapeutic agent and enhances its efficacy (Maeda et al., 2000). By taking advantage of nanoscale phenomena, we are capable of designing carriers that improve the delivery of therapeutic molecules, modulate their release profiles, and respond to specific intra-body stimuli, such as pH or enzymatic activity changes. It is this level of control that leads to greatly improved therapeutic outcomes with minimized side effects and improved patient compliance (Blanco et al., 2015).

1.2 FUNDAMENTALS OF NANOCARRIERS

1.2.1 Definition and Characteristics of Nanocarriers

Nanocarriers can be defined as nanoparticles engineered to deliver therapeutic agents, including nucleic acids and proteins, to target cells or tissues. Nanoparticles possess a diminutive size and extensive surface area, facilitating unique interactions with biological systems, thereby offering an optimal approach to address shortcomings in traditional delivery methods. The performance of nanocarriers in delivering therapeutic nucleic acids and proteins depends on the understanding of their defining characteristics. Certain key attributes that have a bearing on their functionality and delivery include the following:

- *Size*: Nanocarriers are typically 10–100 nm in size. A size of this magnitude allows them to cross various biological barriers and to be internalized by cells through paths that are endocytic in nature, as reported by Blanco et al. (2015). This size influences biodistribution, cellular uptake, and clearance from systemic circulation.
- *Surface Charge*: This is important for interaction with cellular membranes and biological molecules. Positively charged cationic nanocarriers can enhance the delivery of nucleic acids due to enhanced electrostatic interactions with negatively charged cell membranes and nucleic acids (Guo et al., 2024). High positive charge may, however, cause toxicity and non-specific interaction.
- *Surface Chemistry*: The coating of the surface with the polymer, such as PEG, enhances stability and prolongs circulation time due to a lower degree of protein adsorption and lesser recognition by the immune system (Immordino et al., 2006). Targeting ligand(s) can also be attached to the surface for specific delivery to certain cells or tissues.
- *Shape*: While the majority of nanocarriers are spherical, other shapes may include rods or filaments that could influence cellular uptake and eventually biodistribution. Non-spherical particles tend to exhibit prolonged circulation due to their hydrodynamic properties and enhanced accumulation within the target tissue. Functionalization targets molecules like antibodies, peptides, or aptamers that attach with a vector to largely enhance its specificity toward cells or tissues. This improves the therapeutic efficacy of such vectors. An example is given by (Zhang et al., 2015). Functionalization can be done to enable a response against pH or enzyme stimuli to enable controlled release at the target site (Mura et al., 2013).

All of these characteristics have a cumulative effect on how nanocarriers will interact with biological environments, thus determining whether or not they can accomplish their task of delivering therapeutic agents both effectively and safely.

1.2.2 Size-Dependent Properties of Nanoparticles

Size dependence implies that nanoscale materials have different properties from the bulk. Such examples include quantum effects, higher reactivity, and mechanical properties that greatly influence the function of nanoparticle nanocarriers in biomedical applications. One of the most important factors here is the quantum effects emerging in nanosized systems. Because of the dimensions' very small size, electrons become considerably confined, and quantized energy levels are a result, along with changes in photophysical properties, electrical conductivity, and magnetic behavior. For example, quantum dots can exhibit fluorescence whose wavelength is tunable by size, thus becoming excellent tools for bioimaging and diagnostics (Alivisatos, 2003).

The other critical property is the higher surface area-to-volume ratio: as the size of the particles decreases, there is a remarkable increase in surface area relative to volume. Because of this higher surface area, greater interaction with biological molecules is possible, which enhances the loading capacity for therapeutic agents such as nucleic acids and proteins (Cho et al., 2008). This also allows better functionalization of nanoparticles with targeting ligands—for example, polymers or other molecules—which can enhance specificity and stability. Indeed, due to the changed mechanical properties at the nanoscale, nanoparticles have different capabilities in their ability to penetrate physiological barriers intracorporeally. For example, specific

flexibility and deformability enable some nanoparticles to pass through tissues at tight junctions, enabling higher penetration and distribution within the target sites (Discher & Eisenberg, 2002).

Size-dependent cellular uptake is another important factor. Cells take up nanoparticles through mechanisms such as endocytosis. The efficiency of such mechanisms is strongly dependent on the size of the particles. Nanoparticles are generally best for cellular uptake within the range of 10–100 nm. It is a balancing act, because while smaller sizes ensure better internalization, it might also lead to lesser clearance by the immune system (Jiang et al., 2008). Size and shape influence *in vivo* biodistribution and pharmacokinetics: while small nanoparticles deeply penetrate tissues, larger ones may stay longer in circulation.

The size-related properties are also influencing the nanocarrier's ability to bypass the immune system. Nanoparticles can be engineered to avoid recognition by immune cells, thereby prolonging their circulation time and increasing the possibility of reaching the target, as stated by (Zamboni et al., 2012). Size, together with surface properties, determines the way nanoparticles interact with plasma proteins, forming a sort of “protein corona” able to change their biological identity and functionality. Such size-dependent properties can be exploited to design nanocarriers that enable the release of therapeutic agents in controlled ways, responding to specific stimuli, thereby generally improving therapeutic efficacies in drug delivery. It is possible to further develop highly specialized delivery systems for use in targeted medical applications by refining such nanoparticle size and surface properties.

1.2.3 Types of Nanocarriers

1.2.3.1 Organic Nanocarriers

Organic nanovehicles comprise liposomes and polymeric nanoparticles, which are critical in enabling the therapeutic delivery of nucleic acids and proteins. These are typically prepared from biocompatible and biodegradable materials such as lipids and respective polymers. The rationale for engineering design is to improve such agents' stability, bioavailability, and/or targeting ability (Figure 1.1).

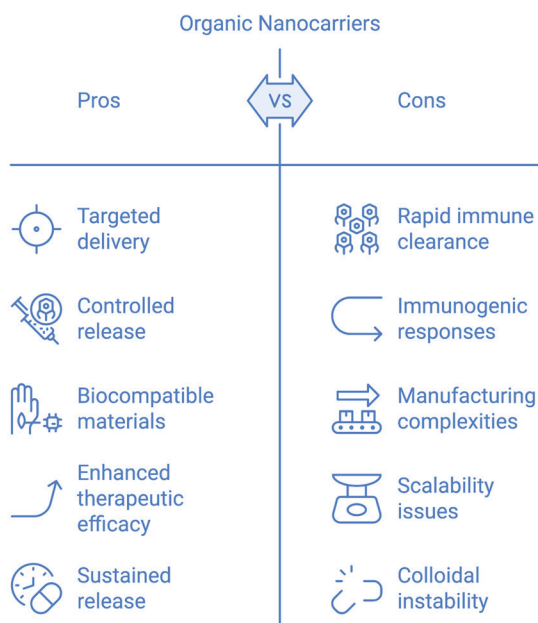


Figure 1.1 Pros and cons of organic nanocarriers in biomedical applications. This figure presents a comparative analysis of the advantages and disadvantages of organic nanocarriers used in drug delivery and therapeutic interventions. Organic nanocarriers, such as liposomes and polymeric nanoparticles, offer targeted delivery, controlled and sustained release, biocompatibility, and enhanced therapeutic efficacy, making them ideal for delivering sensitive molecules like nucleic acids and proteins. However, challenges such as rapid immune clearance, immunogenic responses, manufacturing complexities, scalability issues, and colloidal instability pose significant obstacles to their widespread clinical use.

Liposomes are spherical vesicles with a lipid bilayer, just like cell membranes. Their amphiphilic nature gives them the capacity to encapsulate both hydrophilic and hydrophobic molecules, protecting the sensitive nucleic acids and proteins from degradation through interaction with the biological environment (Allen & Cullis, 2013). Targeting the liposome surface with ligands or polymers ensures specific delivery to particular cells or tissues, allowing for increased therapeutic efficacy with minimal side effects.

Polymeric nanoparticles, made from materials such as PLGA, provide controlled and sustained release of therapeutic agents. Nanoparticles would therefore be able to release cargo at a predetermined rate because they are degradable at a fixed rate over time, which can be critical in therapies using precisely correct dosing over very long periods of time (Danhier et al., 2012). Surface functionalization of polymeric nanoparticles increases cellular uptake and targeting specificity, thereby enhancing the delivery of nucleic acids and proteins to desired sites within the body.

Though organic nanocarriers have their advantages, there are certain limitations. One major drawback is its potential rapid clearance by a part of the immune system called the mononuclear phagocyte system, or MPS, leading to reduced circulation time and less therapeutic effect (Moghimi et al., 2005). Other characteristics that can affect stromal transport impair the colloidal stability of these carriers in the circulation, which, in turn, may lead to premature leakage from the carrier before reaching the target site.

Another drawback involves possible immunogenic responses. Components of liposomes or polymers may be immunogenic, hence limiting the treatment effects by adverse reactions (Immordino et al., 2006). Also, there are manufacturing complexities and scalability issues because, for the production of consistent and reproducible nanocarriers, one has to have precise control of size, composition, and surface characteristics.

1.2.3.2 Inorganic Nanocarriers

Inorganic nanocarriers, with gold nanoparticles and quantum dots as their most important representatives, have unique optical, magnetic, and electrical properties that can be irreplaceable in imaging and therapeutic applications. There are a number of strategies for optimizing these carriers to enhance the delivery and targeting of active therapeutic agents without causing toxicity (Figure 1.2).



Figure 1.2 Pros and cons of inorganic nanocarriers in drug delivery systems. This figure highlights the advantages and challenges associated with inorganic nanocarriers in therapeutic applications. Inorganic nanocarriers, such as gold nanoparticles and quantum dots, offer unique properties including targeted delivery, reduced toxicity, biodegradable options, and customizable dosage, making them valuable in precision medicine. However, they present challenges such as potential toxicity, complex optimization, the need for extensive *in vivo* testing, high production costs, and significant regulatory hurdles that must be addressed for their successful translation into clinical practice.

First, surface modification is critically important. Coating the surface of inorganic nanoparticles with biocompatible materials like PEG reduces their immunogenicity and toxicity, extending the circulation time within the blood (Jokerst et al., 2011). Functionalization of the surface by targeting ligands, such as antibodies or peptides, allows nanocarriers to home to the diseased cells, thereby increasing the therapy's selectivity (Bardhan et al., 2011). Second, the size and shape of the nanoparticles control their biodistribution and cellular uptake. Nanoparticles from 10 to 100 nm can avoid clearance by the MPS effectively and are readily internalized by cells (Jain et al., 2012). Moreover, spherical nanoparticles typically exhibit less toxicity than rod-shaped nanoparticles because of different cellular uptake mechanisms. Third, biodegradable inorganic materials could ensure minimal long-term toxicity. For example, silica nanoparticles can be surface engineered to degrade into harmless silicic acid under physiological conditions; this can reduce the accumulation of nanoparticles in organs and tissues to low levels that may result in potentially harmful side effects (He et al., 2011).

1.2.3.2.1 Optimization of Dosage

The therapeutic efficacy and toxicity can be stabilized by optimizing the quantity of therapeutic agent loaded onto the nanocarriers and managing the release kinetics. Thus, controlling the release kinetics can achieve therapeutic efficacy with minimal toxicity by fine-tuning the amount of therapeutic agent loaded onto the nanocarriers (Khan et al., 2019). This safety profile is further improved by the incorporation of stimuli-responsive features, which trigger the drug release by specific physiological conditions, such as pH or temperature. Finally, in-depth *in vivo* testing and toxicological studies are needed for complete elucidation of interactions between inorganic nanocarriers and the biological systems. Extensive studies will enable the detection of possible side effects and provide insight into how to redesign nanocarriers in response to safety concerns (Nel et al., 2006).

1.2.3.3 Hybrid Nanocarriers

Hybrid nanocarriers are innovative systems made up of organic–inorganic hybrid materials that combine the best of both worlds with the goal of providing advantages in a variety of biomedical applications. Hybrid nanocarriers combine the biocompatibility of the organic component with the functional properties of the inorganic substance, resulting in synergistic delivery of nucleic acids and proteins (Yanar et al., 2023) (Figure 1.3).

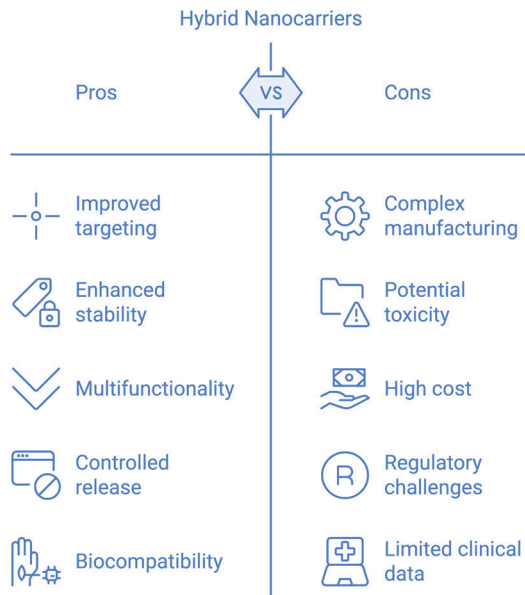


Figure 1.3 Pros and cons of hybrid nanocarriers in biomedical applications. This figure outlines the advantages and challenges associated with hybrid nanocarriers, which combine the best features of organic and inorganic materials. Hybrid nanocarriers offer improved targeting, enhanced stability, multifunctionality, controlled release, and biocompatibility, making them a versatile platform for drug delivery and therapeutic applications. However, the complexity of their manufacturing process, potential toxicity, high cost, regulatory challenges, and limited clinical data present significant barriers to their widespread adoption in clinical practice.

Organic components, such as lipids or polymers, provide a non-toxic and biodegradable matrix that reduces toxicity and enhances the biocompatibility of the system. They enclose active therapeutic agents and reduce the degree of degradation, ensuring also controlled release. Inorganic constituents of the hybrid structure provide added properties, such as gold nanoparticles, quantum dots, or magnetic nanoparticles, which provide optical responsiveness, magnetic resonance, and electrical conductivity (Yetisgin et al., 2020).

The hybrid nanocarriers are capable of the following upon the combination of both:

- **Improve Targeting:** Surface modification by organic ligands allows for specific targeting of particular cells or tissues. Contribute the inorganic component to imaging and tracking, ensuring that the carriers reach their proper destinations.
- **Improve Stability and Efficacy:** By providing structural integrity, the inorganic materials protect nucleic acids and proteins during circulation. This would provide stability, allowing higher concentrations of the therapeutic agent to arrive at the target site.
- **Enable Multifunctionality:** Hybrid carriers can frequently perform both therapeutic and diagnostic functions simultaneously, such as drug delivery, allowing for real-time imaging of their distribution and release.
- **Stimuli-Responsive Release:** Because of the presence of inorganic components, such systems could be made sensitive to external stimuli—light, heat, or magnetic field—ensuring on-demand and site-specific release of therapeutic agents.

These synergies ensure that hybrid nanocarriers avoid problems associated with purely organic or inorganic systems and provide a flexible platform for enabling innovative drug delivery and personalized medicine (Chehelgerdi et al., 2023).

1.2.4 Physicochemical Properties Affecting Nanocarrier Behavior

Among all the physicochemical properties affecting nanocarrier behavior in biological systems, size and shape are the most influential. These properties also determine the nanocarrier's interaction with cells or tissues in general, which in turn determines its biodistribution, cellular uptake, and removal from the body (Figure 1.4).

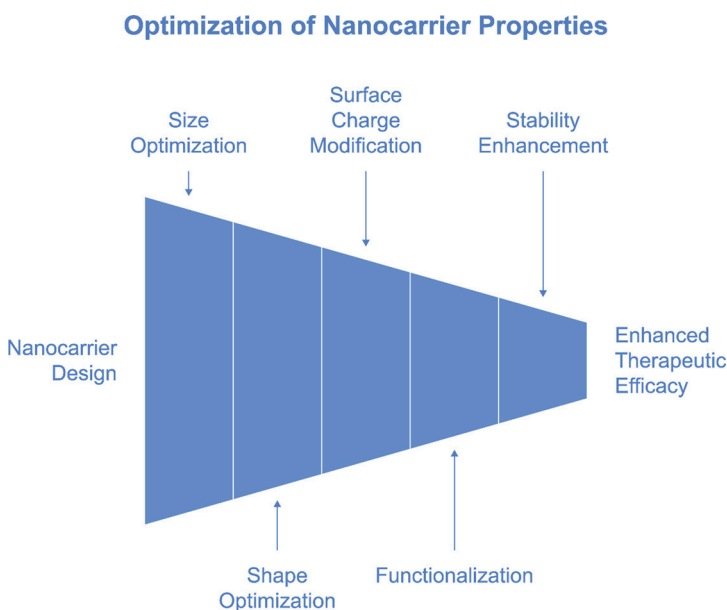


Figure 1.4 Optimization of nanocarrier properties for enhanced therapeutic efficacy. This figure illustrates the critical design factors that contribute to optimizing nanocarriers for drug delivery systems. The design and performance of nanocarriers can be enhanced through size and shape optimization, surface charge modification, functionalization, and stability enhancement. These optimizations lead to improved therapeutic outcomes by enabling better targeting, prolonged circulation, and controlled release of the therapeutic agents, ultimately increasing the overall efficacy of nanocarriers in biomedical applications.

1.2.4.1 Size and Shape

For many aspects described below, the size and shape of nanocarriers are the most critical factors that influence delivery efficiency and therapeutic outcome.

1.2.4.1.1 Biodistribution and Tissue Penetration

- **Size Impact:** From 10 to 100 nm, nanocarriers exhibit the dimensions ideally needed for systemic delivery. Particles below 5 nm in dimension pass out quickly through the kidneys, while particles above 200 nm are cleared through sequestration via the spleen and liver (Blanco et al., 2015). Reduced size can access deep behind tissues and penetrate biological barriers more effectively, thereby reaching target sites more easily.
- **Shape Effect:** The shape of nanocarriers is determined by their ability to navigate through the extracellular matrix and the vasculature. For example, rod-shaped or filamentous particles, compared with their spherical counterparts, have demonstrated enhanced margination and adhesion to vascular walls, thereby enhancing accumulation at target sites (Toy et al., 2014).

1.2.4.1.2 Cellular Uptake Mechanisms

- **Size Dependence:** The nanocarriers are mainly taken up by the cells through a process called endocytosis, which is size-dependent. Particles of about 50 nm are usually internalized more effectively by the cells. Larger particles may not readily be internalized, hence reducing delivery efficiency (Zhang et al., 2015).
- **Shape Dependence:** Generally symmetric, spherical nanoparticles are internalized easily. On the other hand, there are particles of other shapes that have longer circulation times, but at the cost of slower cellular uptake. By design, the shape-engineered nanoparticle takes advantage of specific uptake pathways (Gratton et al., 2008).

1.2.4.1.3 Circulation Time and Clearance

- **Size Effect:** There are intermediate-sized nanocarriers that can evade both rapid renal clearance and uptake by the MPS, allowing them to remain in circulation longer. The extended time in the bloodstream allows more time for the carriers to reach the targeted tissues (Longmire et al., 2008).
- **Shape Factor:** Filamentous or elongated particles were observed to exhibit longer circulation times compared to spherical ones because their geometry impairs recognition and elimination by immune cells (Geng et al., 2007).

1.2.4.1.4 Therapeutic Efficiency

- **EPR Effect:** The tumor tissue's vasculature is typically leaky, allowing nanocarriers of a certain size to accumulate, preferably in tumor tissues. Particles in the nanorange of 30–200 nm have been seen to exploit this EPR effect and improve the drug delivery to tumors (Maeda et al., 2013).
- **Improved Targeting Efficiency:** Optimizing carrier size and shape might help to engineer nanocarriers for enhanced binding affinity toward target cells and tissues, resulting in improved therapeutic efficacy while causing minimal off-target effects (Skowicki et al., 2024).

1.2.4.2 Surface Charge and Functionalization

Therefore, surface modification of nanocarriers is one of the important aspects of improving targeting accuracy and reducing off-target effects in therapeutic delivery. The addition of specific functional groups and surface charge adjustments improves the precise interaction between nanocarriers and target cells/tissues.

1.2.4.2.1 Influence of Surface Charge

The surface charge of nanocarriers has a high impact on their stability, biodistribution, and cell uptake.

- **Cationic Nanocarriers:** Cationic nanocarriers interact well with the negatively charged cell membranes and nucleic acids. Positive charge promotes cellular internalization and facilitates agents like siRNA and DNA. On the other hand, very high positive charge might cause non-specific interaction with healthy cells and proteins, leading to toxicity and immunogenicity (Yang & Luo, 2023).

- **Negative or Neutral Charge:** Though neutral or slightly negative charge on the surface diminishes the non-specific binding to proteins in the blood, it increases the circulation time as well as the stability of nanocarriers (Immordino et al., 2006). It can be done by coating the surface with hydrophilic polymers like PEG, also called PEGylation.

1.2.4.2.2 Functionalization with Targeting Ligands

Functionalization refers to the attachment of specific targeting ligands on the surface of nanocarriers to achieve active targeting.

- **Antibodies and Antibody Fragments:** Because nanocarriers link antibodies that recognize tumor-specific antigens, nanocarrier attachment to cancer cells will occur more selectively, thereby improving targeting accuracy (Nobs et al., 2004).
- **Peptides and Aptamers:** It is possible to design small peptides and nucleic acid aptamers to attach to special receptors overexpressed on diseased cells to enhance cellular uptake via receptor-mediated endocytosis (Zhao et al., 2020).
- **Small Molecules:** Through attachment, small molecules such as folic acid target the folate receptor, which is often overexpressed in many cancers, thereby increasing delivery to those cells carrying such receptors (Lu & Low, 2012).

1.2.4.2.3 Benefits of Surface Modification

- **Improved Specificity:** Functionalized nanocarriers target specific cells. Increasing the local concentration of the therapeutic agent, thereby improving efficacy, is also enhanced (Elumalai et al., 2024).
- **Reduced Off-Target Effects:** There are fewer side effects because of the minimum off-target effects; it spares the healthy cells and hence is normally a better therapeutic index (Niculescu & Grumezescu, 2022).
- **Improved Cellular Uptake:** Targeting ligands provides a means of entry into cells through specific pathways, thereby increasing nucleic acid and protein delivery efficiency (Yan et al., 2024).
- **Stealth Properties:** Surface modifications, such as PEGylation, reduce recognition by the MPS, which can prolong the circulation time and increase the possibility of reaching target tissues (Raza et al., 2017).

1.2.4.2.4 Issues and Considerations

There is supposed to be an optimal balance between charge and toxicity. While cationic surfaces enhance the uptake, they also appear to disrupt the membrane, leading to toxicity. A proper balance should be achieved to maximize delivery with minimum adverse effects (Sanità et al., 2020).

- **Stability of Ligand Attachments:** The linkage between targeting ligands and nanocarriers should be stable in circulation but allow for effective interaction at the target site.
- **Manufacturing Complexity:** Functionalization increases production time due to the need for careful control to ensure regularity and efficacy.

1.2.4.3 Stability and Degradation

The balance between stability and controlled degradation remains extremely critical toward the effectiveness and safety of nanocarriers in therapeutic delivery. While it is important for nanocarriers to be stable in circulation to avoid the release of their payload before time attains destinations, they ought to degrade appropriately to release the therapeutic agents upon reaching their destination.

1.2.4.3.1 Strategies to Optimize Stability and Controlled Degradation

- **Surface Treatment: Application of Protective Coatings:** Coating with hydrophilic polymers, such as PEG, reduces protein adsorption and recognition by the immune system, hence increasing the stability of nanocarriers. This process is described as PEGylation: this prolongs circulation time and minimizes clearance through MPS (Owens & Peppas, 2006).
- **Application of Biodegradable Materials:** The use of a biodegradable polymer such as PLGA in forming nanocarriers provides for controlled degradation. Materials made up of such polymers degrade through

hydrolysis into biocompatible monomers, which pass through metabolism before being excreted by the body. According to Danhier et al. (2012), further fine-tuning of the degradation rate can be achieved through alterations in the composition and molecular weight of the polymer.

1.2.4.3.2 Incorporation of Stimuli-Responsive Elements

The design of nanocarriers that respond to specific physiological stimuli ensures degradation/payload release only under specified conditions:

- **pH-Sensitive Materials:** These are materials that degrade or change configuration due to a difference between pH values of the normal and pathologic tissues, for example, the tumor environment. This allows for active targeting by releasing a payload (Binauld & Stenzel, 2013).
- **Enzyme-Responsive Linkages:** Cleavage of incorporated linkages by enzymes, which are overexpressed in target tissues, results in controlled degradation (Hu et al., 2014).
- **Redox-Sensitive Systems:** The utilization of disulfide bonds, which cleave under reductive conditions of the cytoplasm and enable intracellular release of the payload, has been employed by redox-sensitive systems. This approach has been utilized by Duan et al. (2013).

While in circulation, cross-linking of nanocarrier components improves stability and prevents premature degradation. Cleavable cross-linkers may allow the nanocarrier to remain intact during circulation but degrade at the site of action upon arrival (Kataoka et al., 2012). Nanocarriers with the ability to control composition and thickness prepared by layer-by-layer (LbL) assembly have been constructed for enhanced stability. Such outer layers are engineered to degrade under specific conditions; this facilitates the release of the inner payload (De Koker et al., 2012).

The modulation of size, shape, and surface charge in nanocarriers influences their interaction with biological hosts and stability; for instance, particle size optimization reduces opsonization and clearance, thereby enhancing circulation stability (Blanco et al., 2015). Nanocarriers can be protected by shells or nanocapsules to avoid degradation. Such shells may be made to disassemble under some stimulus conditions, with the ensuing release of their content in a controlled fashion (Torchilin, 2014).

1.3 NANOCARRIERS FOR NUCLEIC ACIDS

Delivering nucleic acids as a therapeutic agent poses considerable problems due to several major *in vivo* barriers. These obstacles include enzymatic degradation by nucleases, large size, and negative charge, all of which lead to a general inability to cross cellular membranes.

1.3.1 Challenges in Nucleic Acid Delivery

Naked DNA and RNA rapidly undergo degradation due to the presence of nuclease enzymes in systemic blood circulation and extracellular fluids. These enzymes cleave the nucleic acids at a rapid rate, rendering them ineffective even before they could penetrate the target cells. According to Mehta et al. (2023), the severe instability of nucleic acids in biological environments greatly restricts their therapeutic potential. Also, the cell membrane is a phospholipid bilayer that acts as a crucial permeation barrier to large molecules with negative charge, including nucleic acids (Kotzé et al., 1999). Due to their phosphate backbone, which imparts a negative charge, they are repelled by the cell surface, which is equally negatively charged, and passive diffusion across the membrane is highly ineffective. As a result, when nucleic acids are administered without a delivery system, they have a low cellular intake. Naked nucleic acids could also trigger an immune response because the body identifies them as noxious agents. Immune activation leads to inflammation that dislodges therapeutic agents from the body, reducing treatment efficacy. In severe cases, such an immune response could have toxic side effects (De Jong et al., 2007).

The problem can be resolved by nanocarriers, which encapsulate nucleic acids and protect them from enzymatic degradation. LNPs physically protect nucleic acids from nuclease activity, increasing their half-life in circulation and augmenting their probabilities of reaching the target cells (Gilleron et al., 2013). Changing their superficial properties promotes cellular internalization, such as endocytosis. For example, cationic nanocarriers, through electrostatic interactions, interact with the negatively charged cell membrane and promote internalization. This has been developed one step further to where the attachment of ligands such as antibodies or peptides on the nanocarriers' surface can target a certain site and induce receptor-mediated endocytosis, which enhances specificity in cellular uptake (Shi et al., 2023; Won et al., 2011). Nanocarriers can neutralize the negative charge of nucleic acids through complexation with positively charged polymers or lipids. It reduces electrostatic repulsion from the cell membrane, allowing for close interaction and intracellular uptake (Zhang et al., 2007). Coating nanocarriers with hydrophilic polymers, such as PEG, can

make them “invisible” to the immune system by avoiding opsonization and eventual phagocytic clearance. This stealth property prolongs circulation time and improves biodistribution (Shi et al., 2021).

Moreover, nanocarriers can be engineered for the delivery of nucleic acids to target cells or tissues with minimal off-target effects. The addition of targeting ligands on nanocarriers imparts the recognition capability and specific binding to receptors overexpressed on diseased cells (Liu et al., 2022). Targeted delivery enhances therapeutic efficacy with reduced systemic toxicity. Intracellularly, nucleic acids have to escape the endosome into either the cytoplasm or nucleus—wherever they exert a therapeutic effect. Nanocarriers can be engineered to contain endosomolytic agents or acid-labile materials that disrupt the endosomal membrane, allowing nucleic acids to be released into the cytoplasm (Ahmad et al., 2019).

1.3.2 Types of Nucleic Acids Used in Therapy

In general, therapeutic nucleic acids include a range of molecules that express different features in structure and function; this is important for defining the better design for their nanocarriers. Currently, the most commonly used therapeutic nucleic acids include siRNA, miRNA, mRNA, and pDNA.

1.3.2.1 Small Interfering RNA

siRNA molecules are small, usually 21–23 nucleotides long, double-stranded RNA. They initiate RNA interference, RNAi, which results in the knockdown of specific genes by degrading complementary mRNA sequences (Elbashir et al., 2001). Because of its small and double-stranded nature, siRNA necessitates nanocarriers to protect it against enzymatic degradation while enabling efficient cellular uptake and escape from endosomes.

1.3.2.1.1 Design Implications for siRNA Nanocarriers

- *Protection against Degradation:* Nanocarriers must protect siRNA from the nucleases present in circulation.
- *Improved Cellular Uptake:* Complexes of cationic lipids or polymers with siRNA can also serve as a delivery vehicle for improved cellular entry.
- *pH-Responsive Materials for Endosomal Escape:* These materials, upon cellular internalization, help in releasing siRNA into the cytosol.

1.3.2.2 MicroRNA

miRNAs are single-stranded, endogenous RNA molecules usually 18–25 nucleotides in length. They regulate gene expression by binding to target mRNAs, leading to translational repression or degradation (Bartel, 2004). miRNAs have therapeutic applications involving the use of miRNA mimics or antagomiRs that modulate the activity of miRNAs in diseases.

1.3.2.2.1 Design Implications for miRNA Nanocarriers

- *Targeted Delivery:* Precision is key to avoiding off-target effects; nanocarriers often incorporate targeting ligands that guarantee miRNA delivery into target cells.
- *Sustained Stability:* miRNA must be shielded from degradation by stable encapsulation or complexation.
- *Controlled Release:* Over time, sustained release profiles help to maintain therapeutic levels.

1.3.2.3 Messenger RNA

mRNA therapies involve the delivery of mRNA sequences that encode therapeutic proteins. Unlike DNA-based therapies, mRNA operates in the cytoplasm and does not integrate into the genome; this reduces potential safety risks associated with its use (Sahin et al., 2014). However, mRNA is relatively large and unstable, making its delivery a challenge.

1.3.2.3.1 Design Implications for mRNA Nanocarriers

- *Size Accommodation:* The nanocarriers must be able to encapsulate large mRNA molecules.
- *Stabilization:* This is done by ionizable lipids or polymers, which protect mRNA from degradation by RNases.

- *Efficient Translation:* Delivery systems should ensure the release of mRNA into the cytoplasm to allow translation to take place.

1.3.2.4 Plasmid DNA

pDNA is double-stranded, circular DNA that is used to deliver genes either for replacement or addition therapies. For this sort of therapy to be effective, pDNA must reach the cell nucleus (Kay et al., 2001).

1.3.2.4.1 Design Implications for pDNA Nanocarriers

- *Nuclear Targeting:* Nuclear localization signals can be implanted on nanocarriers to enhance nucleus transportation.
- *Protection from DNases:* Encapsulation helps protect pDNA against enzymatic degradation.
- *Large Payload Capacity:* Nanocarriers must accommodate the larger pDNA size compared to RNA molecules.

1.3.2.4.2 Influence of Structural and Functional Differences

- *Size and Complexity:* Larger-sized nucleic acids, such as mRNA and pDNA, require nanocarriers with higher loading capacity and robust protection mechanisms.
- *Chemical Stability:* RNA molecules are susceptible to degradation, so carriers that provide better stabilization are required.
- *Cellular Targeting:* siRNA and miRNA act in the cytoplasm, while pDNA must be delivered to the nucleus, necessitating the use of targeting signals.
- *Immunogenicity:* Unmodified mRNA and pDNA are immunogenic, so nanocarriers may need to include components that modulate these effects.

1.3.3 Design Considerations for Nucleic Acid Nanocarriers

In the process of delivering nucleic acids to target cells, several critical design parameters are concerned with the design of appropriate nanocarriers. These parameters ensure that nanocarriers can protect nucleic acids from degradation, achieve efficient cellular uptake, and provide targeted delivery to specific cells or tissues (Figure 1.5).

Nucleic acids are highly susceptible to the degrading action of nucleases within the bloodstream and extracellular environment. Nanocarriers must protect these therapeutic molecules by creating a protective environment. This is achieved by encapsulating nucleic acids into LNPs or polymeric nanoparticles to protect them from enzymatic degradation (Mao et al., 2010). Materials that strongly interact with nucleic acids, such as cationic polymers, stabilize the complexes formed (Hadianamrei & Zhao, 2022).

It is necessary for nucleic acids to be taken up efficiently in cells to express their therapeutic effects. Surface modification of nanocarriers with positive charges increases their interactions with negatively charged cell membranes, thus enhancing cellular uptake (Oh & Park, 2009). The size optimization into a range of 10–200 nm optimizes endocytosis and penetration while minimizing the chances of clearance by the immune system (Blanco et al., 2015). Additionally, modifying nanocarriers with cell-penetrating peptides (CPPs) enhances their intracellular delivery by facilitating translocation across cell membranes (Presente & F. Dowdy, 2013).

Targeted delivery is of the utmost importance for enhancing therapeutic efficacy while reducing side effects. Nanocarriers can be functionalized with targeting molecules—such as antibodies, aptamers, or small molecules—that direct them to receptors on target cells (Kwon et al., 2008). Engineering nanocarriers to release their therapeutic payload in response to specific stimuli, such as pH or enzymes present in diseased tissues, further enhances targeting accuracy (Lee et al., 2003). PEG modification of nanocarrier surfaces can reduce non-specific uptake by the immune system, further extending circulation time and increasing the chances of reaching target tissues (Moghimi et al., 2001).

Within the cell, many nucleic acids are sequestered in endosomes prior to their release into the cytoplasm by the nanocarriers. The use of materials sensitive to the acidic environment of the endosome can cause membrane disruption and nucleic acid release (Varkouhi et al., 2010). For example, the proton sponge effect caused by polymers like polyethylenimine (PEI) provokes endosomal swelling and rupture, allowing successful delivery to the cytoplasm (Benjaminsen et al., 2012).

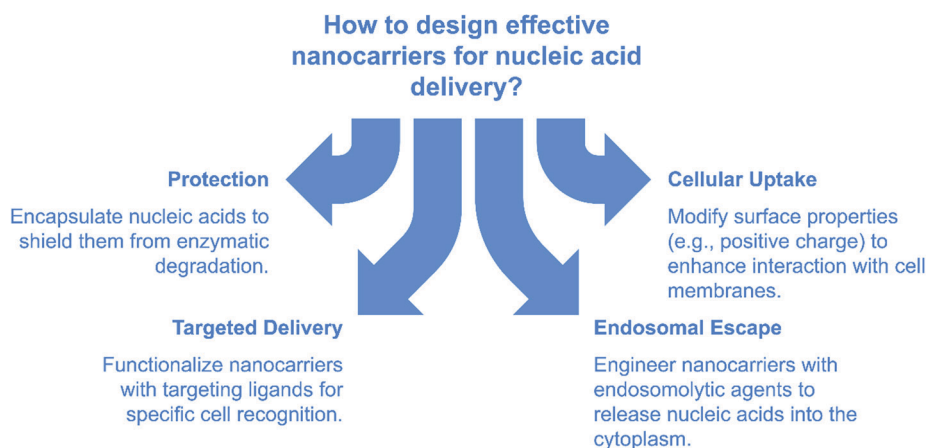


Figure 1.5 Key design strategies for effective nanocarriers in nucleic acid delivery. This figure illustrates four essential strategies for designing nanocarriers that optimize the delivery of nucleic acids, which include (1) protection, by encapsulating nucleic acids to shield them from enzymatic degradation; (2) cellular uptake, achieved through surface modifications (e.g., positive charge) to enhance interactions with cell membranes; (3) targeted delivery, where nanocarriers are functionalized with targeting ligands for specific cell recognition; and (4) endosomal escape, by engineering nanocarriers with endosomolytic agents that facilitate the release of nucleic acids into the cytoplasm after internalization. These strategies collectively improve the efficacy of nucleic acid therapies by overcoming biological barriers.

To achieve safety and tolerability, nanocarriers should be designed with biodegradable materials that degrade into non-toxic byproducts, minimizing long-term toxicity (Danhier et al., 2012). Additionally, avoiding components that may be recognized by the immune system further enhances the biocompatibility of nanocarriers, reducing unwanted adverse reactions (Fan et al., 2023).

Nanocarriers developed for clinical applications must be manufacturable at scale with guaranteed quality (Patel et al., 2021). Robust synthesis methods ensure reproducibility in batches, while strict quality control protocols help maintain the integrity of nanocarrier formulations throughout production (Buya et al., 2024).

1.3.4 Examples of Successful Nucleic Acid Nanocarriers

The enormous success of LNPs with mRNA vaccines, and in particular, those from Pfizer-BioNTech and Moderna against COVID-19, has shown the transforming power of nanocarriers in the clinic. These vaccines have demonstrated one way through which LNPs can successfully deliver nucleic acids to achieve high immune responses and very high efficacy rates (Baden et al., 2020; Polack et al., 2020). There were a number of factors that contributed to the clinical success of LNPs. One such factor was efficient delivery and uptake. LNPs are designed so that mRNA molecules are located within the core, quite well protected against nuclease-mediated degradation in the bloodstream. Such a design allows for the effective uptake of LNPs by host cells through endocytosis and the concomitant access of mRNA to the cytoplasm, where translation into the desired protein occurs (Hassett et al., 2019).

Improvement in stability and protection is also an important factor. The lipid components of LNPs, especially the ionizable lipids, form stable complexes with negatively charged mRNA, protecting its integrity during storage, handling, and administration of the concocted vaccine preparation (Hou et al., 2021). While these vaccine types are administered intramuscularly, LNPs are specifically tailored to target the antigen-presenting cells in the local tissue to enhance the immune response. PEG is incorporated on the surface of the nanoparticles, serving to prolong the circulation time and reduce non-specific interactions (Samaridou et al., 2020).

Scalability and manufacturability have formed another important attribute to their success. The manufacturing processes of the LNPs were optimized for large-scale productions, which has been a crucial determinant in ensuring that supplies meet the global demand for the vaccines. For example, methods such as microfluidic mixing allow nanoparticles to be generated consistently and in a reproducible manner (Cullis & Hope, 2017). Preclinical and clinical trials have also established safety and tolerability. LNP-formulated mRNA vaccines have demonstrated an acceptable safety profile, with the majority of the side effects being ranked as mild to moderate. The use of biodegradable and biocompatible lipids minimizes long-term toxicity

problems (Jackson et al., 2020). This success underlines several perspectives that could be useful in the design of future nanocarriers. For instance, the chemical tailoring of the lipid components allows perspectives in enhancing delivery to target tissues or cell types.

Design improvements of ionizable lipids have enabled improved endosomal escape and reduced the required dose (Sun & Lu, 2023). Besides, the placing of targeting ligands on the surface of LNPs would redirect them to target cells, like cancerous cells or cells within tissues that are hardly reached, improving their therapeutic outcome (Safari & Zarnegar, 2013). Other areas of interest involve improvements in mRNA stability. Modifications to the mRNA itself, such as the incorporation of nucleotide analogs, can improve stability and promote translation at lower doses, with less immune activation versus the mRNA itself, while modifications to the delivery vehicle can be used to limit off-target effects (Pardi et al., 2018). Improvements in manufacturing techniques include the development of scalable and cost-effective methods, including continuous production, which will be important for the broader adoption of nanocarrier-based therapies in general (Su & Kang, 2020). In addition, there is still one challenge to be overcome: immunogenicity. Research on the immunogenicity of nanocarriers and their components will make possible the design of systems in ways to reduce unwanted immune responses, thus improving the safety profile of nanosized drug delivery systems (Bahl et al., 2017).

1.3.5 Mechanisms of Cellular Uptake and Endosomal Escape

The entrance of nanocarriers into cells and the release of their cargo is pivotal in effective nucleic acid delivery. Two major steps—enhancement of cellular uptake and facilitating endosomal escape—play an important role in the success of these delivery systems.

Different strategies can be developed to enhance cellular uptake for nanocarriers. Surface modification with targeting ligands is considered one of them. Receptor-mediated endocytosis could be facilitated by the attachment of ligands such as antibodies, peptides, or aptamers onto the nanocarrier surface. This method enables nanocarriers to recognize and bind receptors that are overexpressed on target cells, hence increasing specificity (Kwon et al., 2008). Other factors that are crucial in enhancing uptake include particle size and shape optimization. Nanocarriers designed within the optimal size range, usually between 50 and 200 nm, are internalized easily by cellular elements. Moreover, the shape of the nanocarrier could affect internalization pathways—for example, rod-shaped particles may be internalized differently compared to spherical ones (Gratton et al., 2008). Another approach is the modification of the surface charge; slightly positively charged nanocarriers interact more effectively with the negatively charged cell membrane, thus improving the uptake of the particle. However, excess positive charge can induce toxicity; thus, a proper balance has to be maintained (Oh & Park, 2009). The decoration of nanocarriers with CPPs is another strategy that promotes nucleic acid intracellular delivery by allowing nanocomplexes to translocate through cellular membranes via energy-independent pathways, which further promotes the intracellular delivery of nucleic acids (Nakase et al., 2012).

Intracellularly, nanocarriers are often trapped in endosomes and need strategies for efficient release of nucleic acids into the cytoplasm. One method includes pH-sensitive materials. These materials could be collectively designed to respond in the acidic environment of the endosomes at pH ~ 5–6 that might result in structural changes of the nanocarrier, disruption of the endosomal membrane, and finally release of the cargo (Du et al., 2015). Another potent mechanism is the proton sponge effect, where one could incorporate appropriate amounts of polymers such as PEI (Benjaminsen et al., 2012). These are the polymers that buffer the acidified environment of the endosome, leading to an influx of ions and water into the polymer particles. This influx causes the polymer particles to swell and rupture the endosome. Membrane-disruptive peptides, which become active under acidic conditions, are also employed to disrupt the endosomal membrane and thereby easily release nucleic acids into the cytoplasm (Bechinger & Lohner, 2006). Another sophisticated strategy involves photochemical internalization, which employs light-sensitive compounds to trigger the activation of nanocarriers. These compounds force endosomal escape upon irradiation with light and allow for precise control of the release process (Soe et al., 2020). These approaches might be combined in such a way that it would be possible to engineer nanocarriers to be optimal for both cellular uptake and endosomal escape.

Targeting ligand, pH-responsive material, and endosomal escape agent combined into one multifunctional nanocarrier—an approach that improves the overall nucleic acid delivery efficiency (Pandey et al., 2024). Nanocontainers prepared by an LbL assembly approach offer excellent capabilities for the precise control of surface properties and functional components of these assemblies toward the optimization of targeting and release profiles (De Koker et al., 2012). For instance, a stimuli-responsive system could be designed to be sensitive to specific intracellular conditions, such as reductive environments or enzymatic activities, under which the nucleic acids are released only within target cells. This concept has been demonstrated by Du et al. (2015). These carefully engineered strategies allow nanocarriers to overcome biological barriers, thus

ensuring that nucleic acids are effectively delivered into the desired cells while maximizing the therapeutic potential with minimal unwanted side effects.

1.4 NANOCARRIERS FOR PROTEINS

In terms of delivery challenges, proteins are unique therapeutic agents due to their complexity and extreme sensitivity. Maintaining biological activity and ensuring stability during the delivery process are essential for their efficacy.

1.4.1 Challenges in Protein Delivery

There are numerous significant challenges associated with protein delivery. Of those, the most important is denaturation and degradation. Due to changes in temperature, pH, and other environmental problems, proteins can lose their three-dimensional (3D) structure—a process called denaturation. Moreover, therapeutic proteins have a very short-lived half-life in the body due to enzymatic digestion by proteases, which often makes the loss of their functional level possible (Wang et al., 2006). The other challenge is immunogenicity as several therapeutic proteins have a tendency to activate immune responses. This could lead to the generation of neutralizing antibodies, which interfere either with the efficacy of proteins or lead to adverse immune responses (Hermeling et al., 2004). Another challenge arises due to poor cellular uptake. Protection from accessing intracellular targets occurs by virtue of poor permeability through cellular membranes due to their large size and hydrophilic nature (Haddadzadegan et al., 2022). The circulating half-life is rather short for many proteins because of rapid clearance by the reticuloendothelial system (RES), thus further narrowing their therapeutic window (Shirian et al., 2024).

Nanocarriers have come to offer innovative solutions to these challenges. Another one of the key advantages nanocarriers possess is protecting proteins against denaturation and degradation. This means that once a protein is encapsulated within the nanocarriers, it becomes shielded from harsh conditions in the environment as well as enzymatic assault. In this regard, the lipid nature of nanocarriers, represented by liposomes and solid LNPs, provides shielding for proteins—per se, a coating composed of a double layer of lipids—that may contribute to the stabilization of proteins. Massignan et al. (2023) have presented the role of nanocarriers in enhancing cellular uptake. The nanoparticles, which are polymeric in nature, and the nanoemulsions are designed to be endocytosis-supportive with distinct shapes. This enables proteins to more effectively reach intracellular targets (Elsababy & Wooley, 2012).

Nanocarriers can also reduce immunogenicity through surface modification with PEG, a process termed PEGylation. This process protects the protein from the immune system, reducing its immunogenicity and allowing it to circulate longer in the bloodstream (Veronese & Pasut, 2005). Another key feature of the nanocarriers is the controlled release. Nanocarriers can be engineered for the controlled and sustained delivery of proteins, maintaining therapeutic levels over extended periods. Some nanocarriers may be stimuli-sensitive; these are designed to release their protein cargo in response to specific triggers, such as changes in pH or enzymes present in target tissues (Ortiz et al., 2024). Nanocarriers can also be functionalized with different targeting ligands, such as antibodies or peptides, to be directed toward a specific needy cell or tissue. This increases therapeutic efficacy while off-target effects are minimized (Todaro et al., 2023). Applying nanocarriers provides solutions to major challenges in protein delivery, such as stability, bioavailability, and efficacy, considerably improving protein therapeutics. This indeed opens ways for the development of more effective treatments in a wide circle of medical fields.

1.4.2 Types of Therapeutic Proteins

The concept of therapeutic proteins enlists a wide variety of molecules with different functions that shape the design requirements for nanocarriers. Among the major classes are enzymes, antibodies, and growth factors; all perform specific functions in treatments.

Enzymes are the biological catalysts for the chemical reactions that take place in the human body. They are administered as medical therapy to take the place of missing or malfunctioning enzymes, for example, in enzyme replacement therapies in diseases such as Gaucher's disease (Desnick & Schuchman, 2002). In the development of nanocarriers for enzymes, several critical factors have to be taken into account. Firstly, the activity of the enzyme is paramount as enzymes are generally sensitive to environmental conditions, which may lead to their denaturation, thereby losing their activity. Nanocarriers need to provide a stable environment that maintains the 3D structure of the enzyme and hence its catalytic activity (Yu et al., 2015). A very important feature is controlled release, as enzymes often need to be delivered at specific sites within the body. This means nanocarriers should be able to release their enzyme payload upon receiving a stimulus like pH changes or enzymatic triggers in target tissues (Mizukami et al., 2017). In addition, most of the therapeutic enzymes are of non-human origin, which may evoke immune responses. Surface modification

of nanocarriers with biocompatible polymers may act to reduce the immunogenicity of nanocarriers (Zhang et al., 2016).

Another major class of therapeutic proteins includes antibodies, which are integral components of the immune system designed to recognize and bind to specific antigens. They have wide applications in treatments against cancer, autoimmune diseases, and infections (Ecker et al., 2014).

Consequently, the design of nanocarriers for antibody delivery becomes rather challenging. It is necessary to keep the correct orientation of the antibodies and their binding potential because their antigen-binding sites need to be free and available. Some examples of immobilizing antibodies by using nanocarriers were given without blocking their binding capability with their respective target antigens. Another critical issue related to the stability of antibodies is that they tend to aggregate and degrade (Chames et al., 2009). In this regard, nanocarriers should provide a microenvironment that protects the antibodies' stability. As stated by Shire et al. (2004), this will serve as an additional benefit. Also, the use of antibodies may enhance targeted delivery (Shire et al., 2004). Conjugated to the surface of nanocarriers, they direct the carrier toward specific cells or tissues, hence increasing specificity and improving cellular uptake (Zhao et al., 2020).

Growth factors are signaling proteins that, through binding with specific membrane-bound receptors, regulate cell proliferation, differentiation, and survival; hence, growth factors are widely used in regenerative medicine and tissue repair (Anitua et al., 2011). Nanocarriers designed for growth factor delivery should protect such sensitive proteins from degradation mediated by proteolytic enzymes. Entrapment of growth factors into nanocarriers protects them from enzymatic degradation. This becomes crucial in the case of growth factors because they are part of a complex signaling process (Yu et al., 2015). Engineering of nanocarriers with the capability to provide sustained or sequential release to be able to mimic natural release patterns of these proteins. The materials used for nanocarriers also have to be biocompatible. In tissue engineering, for instance, they need to support cell adhesion and cell proliferation (Briquez et al., 2015).

Designing nanocarriers for therapeutic proteins needs modification based on the specific role played by those proteins, including stability and release profile. For enzymes and antibodies, especially, stability during transport is critical. As a matter of fact, recent studies have focused on self-assembling protein nanocarriers that could enhance the bioavailability while preserving the structure of therapeutic proteins (Li & Champion, 2022). Stimuli-responsive systems, for instance, release their cargo according to environmental signals such as pH or enzymes. This represents a great leap forward in optimizing therapeutic effects for growth factors, whose activities are finely tuned by both concentration and timing (Qin et al., 2019). Antibodies, on the other hand, often play dual roles both as targeting agents and as therapeutics and thus impose additional multifunctional requirements on nanocarriers at no compromise to efficacy during the delivery process (Knauer et al., 2019). Finally, proteins that might induce some immune effects also require nanocarriers with the capability to camouflage or regulate the immune responses to reduce side effects, as was previously seen in drug delivery with protein-based nanostructures (Krishnan, 2020).

1.4.3 Design Considerations for Protein Nanocarriers

Nanocarriers for protein delivery should be developed by strategies that ensure that protein structure is maintained, rates of release are effective, and targeting of tissues occurs. Proteins are fragile molecules; they may lose their functions by denaturation, degradation, or aggregation. Thus, maintaining stability and bioactivity throughout the process of delivery is a challenge (Muslykhah et al., 2024) (Figure 1.6).

Among these encapsulation techniques, proteins are protected, and their stability and bioactivity are maintained. Proteins encapsulated within the nanocarriers, such as within liposomes, polymeric nanoparticles, or nanogels, are protected against severe physiological conditions and enzymatic degradation that may cause denaturation through changes in their native conformation. Knauer et al. (2019) illustrated that mild preparation methods include self-assembly and electrostatic complexation, which avoid extreme conditions of pH, temperature, or organic solvents, which in turn normally destabilize protein structure (Knauer et al., 2019).

Surface modification and functionalization also play a critical role. Coating nanocarriers with PEG—a process known as PEGylation—improves their stability by preventing protein adsorption and reducing immune recognition. In general, this prolongs the circulation time and diminishes immunogenicity (Veronese & Pasut, 2005). The attachment of targeting ligands to the surface of nanocarriers enables one to achieve specific delivery into target tissues, increasing therapeutic efficacy while further minimizing off-target effects (Zhao et al., 2020).

This precision in protein delivery is further imparted with controlled release mechanisms. The stimuli-responsive nanocarriers undergo environmental pH changes, temperature fluctuations, or enzymatic action to release at the right site and keep the proteins in an inactive state until reaching their intended site. Biodegradable polymers, such as PLA, allow for sustained release—often maintaining therapeutic levels of proteins for weeks—and hence decreasing dosing frequency (Liu et al., 2024).

Enhancing Protein Delivery with Nanocarriers



Figure 1.6 Strategies to enhance protein delivery using nanocarriers. This figure summarizes key approaches to improve the delivery of therapeutic proteins through nanocarriers. These strategies include (1) identifying delivery challenges, (2) utilizing nanocarriers to protect proteins from degradation, (3) enhancing cellular uptake, (4) reducing immunogenicity, (5) enabling controlled and sustained release, (6) functionalizing nanocarriers for targeted delivery, (7) designing nanocarriers for specific protein structures, and (8) implementing self-assembly and surface modifications to ensure stability and bioactivity. These approaches collectively enhance the efficacy and precision of protein-based therapies.

Generally, to stabilize protein structure during formulation and delivery, the addition of stabilizing agents, such as trehalose or other sugars, amino acids, or polymers, to the formulation helps in maintaining the protein tertiary structure (Olgenblum et al., 2024). Proteins freeze-dried within nanocarriers can have improved stability during storage and transportation while retaining bioactivity up to the time of administration (Tchessalov et al., 2023).

Further, immunogenicity is minimized. Encapsulation and surface modification can mask the immunogenic regions of the protein against recognition, thereby causing minimal immune response (Harris & Cohen, 2024). Besides, construction with biocompatible materials minimizes unwanted side effects and intends to keep the functionality of a protein upon delivery (Mitchell et al., 2020).

Another important determinant of successful protein delivery is the ability to optimize nanocarrier properties, such as size and surface charge. While smaller particles generally may penetrate better through tissues, the surface charge can confer different interactions with cell membranes (Blanco et al., 2015). It is also important that the design of soft nanocarriers avoids protein aggregation. This may be achieved by controlling protein loading density and incorporating anti-aggregation agents to decrease protein–protein interactions (Mahler et al., 2008).

Another important aspect is the improvement of cellular uptake. As an example, CPPs attached to nanocarriers enable the effective transport across cellular membranes due to their properties for enhancing intracellular delivery, as observed by Nakase et al. (2012). Another way by which nanocarriers could be modified is by targeting ligands that attach to certain receptors on the cell surface. In this instance, receptor-mediated endocytosis is favored, and many more cells are taken up, increasing delivery efficiency, as stated by Rojekar et al. (2024).

1.4.4 Examples of Successful Protein Nanocarriers

Nanocarriers, including polymeric micelles and nanogels, have exhibited remarkable efficacy in the delivery of therapeutic proteins in both preclinical and clinical investigations. These developments provide essential insights for the design of future protein nanocarrier systems, hence improving their efficacy in many therapeutic applications.

A primary problem in protein distribution is maintaining the stability and bioactivity of proteins throughout the procedure. Nanocarriers mitigate this issue by establishing a protective milieu that safeguards proteins against degradation and denaturation. An exemplary instance is the application of polymeric micelles for enzyme administration. Micelles composed of amphiphilic block copolymers have effectively facilitated the delivery of enzymes. This method involves encapsulating the enzyme within the hydrophobic core of the micelle, safeguarding it from proteolytic degradation, while the hydrophilic shell enhances solubility and extends circulation duration. Research, like that conducted by Kim et al. (2001), demonstrates that this approach improves enzyme stability and increases treatment efficacy in oxidative stress models. This example demonstrates how nanocarriers can maintain protein bioactivity by protecting them from detrimental circumstances during transport.

An additional crucial element is the attainment of controlled and sustained release of therapeutic proteins, which aids in maintaining effective concentrations throughout time and diminishes the necessity for frequent dosage. Nanogels—cross-linked hydrophilic polymer networks—have been utilized for the delivery of growth hormones such as insulin-like growth factor-1 (IGF-1). These nanogels facilitate the prolonged release of IGF-1, enhancing tissue regeneration and repair in myocardial infarction models (Xu et al., 2024). This example illustrates the capability of nanocarriers to improve therapeutic efficacy by sustaining stable protein concentrations at the target location over prolonged durations.

Moreover, precise administration and minimizing immunogenicity are essential for enhancing the safety and efficacy of protein-based treatments. PEGylated liposomes have been utilized to provide interferon-alpha, a therapeutic agent for viral infections and certain malignancies. The use of PEG in these liposomal formulations diminishes immunogenicity and prolongs circulation duration by reducing immune system identification and clearance (Haghighat et al., 2024). This approach enhances therapeutic effectiveness while reducing adverse effects. Surface modification of biocompatible polymers such as PEG enhances targeting accuracy and diminishes adverse immune reactions, resulting in improved patient outcomes.

For proteins that must operate within cells, efficient cellular uptake and endosomal escape are crucial. Nanocarriers enhanced with CPPs have been employed to transport therapeutic proteins, including caspase-3, into cancer cells. CPPs facilitate cellular absorption and enable the release of proteins from endosomes, hence permitting therapeutic proteins to trigger apoptosis in cancer cells (Farkhani et al., 2014). This method illustrates how the functionalization of nanocarriers with CPPs or analogous compounds might enhance intracellular transport, hence broadening the possibilities of protein-based therapeutics.

Moreover, responsiveness to environmental stimuli might augment the accuracy and efficacy of protein delivery. For instance, pH-responsive nanoparticles have been engineered to release anticancer therapeutics exclusively within the acidic microenvironment of malignancies. This tailored release enhances the concentration of the therapeutics at the tumor location while decreasing systemic exposure and mitigating negative effects (Xu et al., 2024). This study demonstrates how stimuli-responsive nanocarriers provide targeted and controlled release in reaction to particular physiological situations, enhancing specificity and therapeutic effectiveness, which could be eventually implemented for therapeutic proteins.

In anticipation of the future, many ways can be employed to enhance protein nanocarrier systems. Primarily, maintaining protein stability will be a concern, with encapsulating methods and material selection being crucial in preventing degradation and sustaining bioactivity. Furthermore, integrating targeting methods, such as functionalizing nanocarriers with antibodies or peptides, would facilitate precise distribution to targeted tissues or cells, hence enhancing therapeutic efficacy. Developing nanocarriers for sustained or stimuli-responsive release, contingent upon the therapeutic protein and disease setting, is crucial for optimizing release patterns.

Enhancing intracellular delivery can be achieved by altering nanocarriers with CPPs or analogous compounds to boost cellular absorption and facilitate endosomal escape. Employing biocompatible and biodegradable materials will reduce toxicity and unfavorable immunological responses, hence enhancing overall patient safety. Ultimately, establishing scalable manufacturing techniques and guaranteeing reproducibility will be essential for transferring these technologies into clinical applications and complying with regulatory norms.

Integrating these insights into forthcoming protein nanocarrier research may yield more efficacious, safer medicines and improved patient outcomes across various medical domains.

1.5 APPLICATIONS IN BIOMEDICAL RESEARCH

Nanocarriers have revolutionized *in vitro* diagnostics by markedly improving the sensitivity and specificity of diagnostic assays. Their capacity to enhance signals, adhere to targets with exceptional precision, and provide swift detection has resulted in the creation of more precise and effective diagnostic assays (Figure 1.7).

Applications of Nanocarriers in Biomedical Research

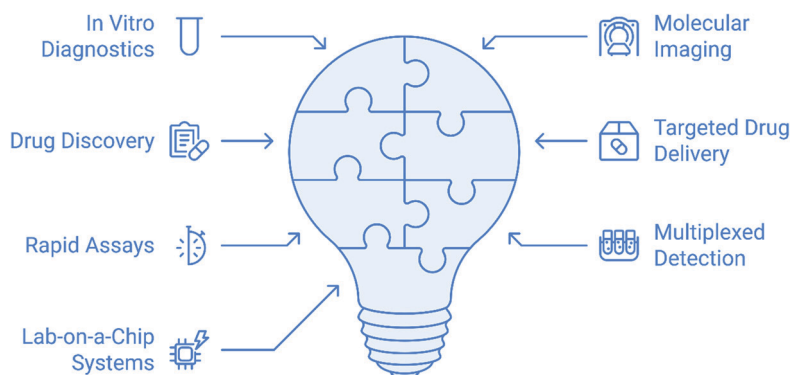


Figure 1.7 Applications of nanocarriers in biomedical research. This figure highlights the wide-ranging applications of nanocarriers in the field of biomedical research. Nanocarriers are being utilized in various areas, including *in vitro* diagnostics, molecular imaging, drug discovery, targeted drug delivery, rapid assays, multiplexed detection, and lab-on-a-chip systems. These applications are pivotal in enhancing the sensitivity, specificity, and efficacy of biomedical interventions, helping to advance precision medicine and improve therapeutic outcomes.

1.5.1 Molecular Imaging and Diagnostics

Nanocarriers enhance the sensitivity of diagnostic tests owing to their elevated surface area-to-volume ratio, adjustable surface chemistries, and capacity for biomolecular functionalization. These attributes enable the amplification of detection signals, enhancement of target selectivity, and acceleration of assay kinetics, rendering them optimal for swift and accurate diagnostics (Mendes et al., 2022).

Nanocarriers primarily boost diagnostic sensitivity by means of signal amplification. Nanoparticles, including gold, quantum dots, and magnetic particles, can transport several signal-generating molecules or have inherent optical characteristics that amplify detection signals. Quantum dots, semiconductor nanoparticles with adjustable fluorescence, are recognized for their exceptional brightness and photostability. These characteristics render them very proficient at identifying low-abundance biomarkers. Quantum dots, when coupled to antibodies or nucleic acids, have markedly enhanced sensitivity relative to conventional fluorophores (Alivisatos, 2003).

Nanocarriers can be modified with ligands such as antibodies, aptamers, or peptides that selectively bind to target analytes. This focused strategy minimizes non-specific interactions, thereby diminishing background noise and enhancing the specificity of diagnostic assessments. Magnetic nanoparticles can be coated with antibodies that selectively bind and concentrate target molecules from intricate biological samples, thereby greatly improving detection limits and enabling the identification of uncommon biomarkers (Khizar et al., 2022).

Due to their nanoscale dimensions, nanocarriers demonstrate enhanced diffusivity, hence expediting their interactions with target analytes. This enhanced efficiency reduces incubation durations, accelerating the total diagnostic procedure. Rapid assays are crucial in point-of-care environments, where prompt results are vital for timely clinical decision-making. Nanoparticle-based lateral flow tests can rapidly identify disease markers, facilitating real-time decision-making and enhancing patient care (Lou et al., 2022).

Nanocarriers facilitate multiplexed detection, permitting the concurrent investigation of numerous biomarkers within a single test. Diverse nanocarriers can be labeled with unique optical or magnetic signatures, enabling the simultaneous detection of several targets. The multiplexing capability is essential for thorough diagnostics, including the profiling of numerous disease indicators to enhance diagnostic precision (Geiss et al., 2008).

Encapsulating sensitive biomolecules in nanocarriers safeguards them against environmental degradation, hence improving the stability and shelf life of diagnostic reagents. This safeguard guarantees uniform assay performance throughout time, rendering nanocarriers optimal for prolonged diagnostic testing. Gold nanoparticles exhibit colorimetric changes following aggregation, facilitating rapid visual identification of target analytes without the necessity for intricate instrumentation (Saha et al., 2012).

Magnetic nanoparticles can be controlled with magnetic fields to effectively isolate and concentrate target molecules, enhancing the sensitivity and rapidity of immunoassays (Khizar et al., 2022). Quantum dots and dye-doped nanoparticles provide intense, sustained fluorescence signals, rendering them exceptionally efficient for identifying low-abundance targets in assays like fluorescence in situ hybridization (FISH) (Alivisatos, 2003).

The integration of nanocarriers with microfluidic devices has facilitated the development of lab-on-a-chip systems that perform intricate diagnostic procedures on small platforms. This integration facilitates automated, high-throughput analysis utilizing minimal sample amounts, hence enhancing the speed and ease of *in vitro* diagnostics (Behera et al., 2023).

Nanocarriers are essential for enhancing *in vivo* imaging by acting as contrast agents in several imaging modalities, including magnetic resonance imaging (MRI), fluorescence imaging, positron emission tomography (PET), and computed tomography (CT). These carriers improve disease detection by augmenting the contrast between healthy and diseased tissues, resulting in more precise and accurate imaging.

Nanocarriers typically measure 10–100 nm in diameter, a dimension that facilitates their effective circulation in the bloodstream, evasion of swift clearance by the RES, and targeting to tissues via the increased permeability and retention (EPR) effect. The size is crucial for optimizing circulation duration and enhancing imaging precision (Blanco et al., 2015). The form also influences performance. Spherical nanocarriers are prevalent due to their straightforward production and advantageous distribution; however, rod-shaped carriers may exhibit extended circulation durations and enhanced targeting capabilities (Geng et al., 2007).

Nanocarriers are frequently coated with hydrophilic polymers such as PEG to diminish immune recognition and prolong their circulation duration (Owens & Peppas, 2006). Surface functionalization with targeting ligands, such as antibodies or peptides, facilitates accurate delivery to targeted tissues, enhancing imaging specificity by ensuring contrast chemicals accumulate in sick regions (Niculescu & Grumezescu, 2022). Efficient nanocarriers must contain or bind imaging chemicals, such as fluorescent dyes, radioactive isotopes, or contrast-enhancing metals such as gadolinium. These medicines must maintain stability within the nanocarrier to prevent premature release and provide dependable delivery to the target region, which is essential for consistent imaging outcomes (Liu et al., 2007).

Nanocarriers composed of biocompatible, biodegradable materials reduce toxicity and immunological responses. Upon completion of imaging, the carriers are expected to decompose into innocuous byproducts, thus mitigating the possibility of long-term buildup and adverse effects (Lei et al., 2022). Nanocarriers that transport numerous imaging agents can markedly enhance signal sensitivity, facilitating disease detection. Certain nanocarriers are engineered to activate their agents only at the target site, hence improving contrast and reducing background noise, resulting in crisper imaging outcomes (Miao et al., 2017).

Nanocarriers can be designed to integrate various imaging agents, facilitating multimodal imaging. A solitary nanocarrier could amalgamate both MRI and PET agents, providing a more holistic diagnostic viewpoint (Lee et al., 2011). Theranostic nanocarriers provide concurrent diagnosis and treatment (Janib et al., 2010). To preserve image quality, nanocarriers must exhibit stability within the body. This entails inhibiting aggregation in the circulation and resisting enzymatic breakdown until the target site is reached (Mura et al., 2013).

Superparamagnetic iron oxide nanoparticles (SPIONs) improve MRI contrast by modifying the relaxation periods of adjacent protons. Coating these particles with polymers and targeting ligands increases their stability and specificity, hence enhancing the detection of malignancies and inflammatory regions (Shubayev et al., 2009). Quantum dots are optimal for fluorescence imaging because of their luminosity and photostability. When modified with targeting molecules, they can identify cancer cells for high-resolution tumor imaging (Gao et al., 2004).

Gold nanoparticles, due to their elevated atomic number, function as efficient contrast agents for CT imaging. Their alteration for biocompatibility improves the imaging of vascular structures and malignancies, providing sharper diagnostic images (Hainfeld et al., 2004). Hybrid nanocarriers merge organic and inorganic materials to facilitate multimodal imaging, such as the combination of optical and PET imaging within a single platform, hence providing a more cohesive diagnostic methodology (Lee et al., 2011).

1.5.2 Drug Discovery and Development

Nanocarriers greatly enhance drug discovery and development by boosting high-throughput screening (HTS), enabling targeted drug administration, and increasing the efficiency of preclinical and clinical investigations. Their distinctive characteristics expedite the identification of promising therapies while enhancing the assessment and safety of potential pharmaceuticals.

Nanocarriers have transformed HTS, an essential phase in drug discovery. Researchers may now simultaneously test hundreds of chemicals by facilitating the creation of nanoarray-based systems. These

technologies immobilize nanocarriers containing potential medications onto microarrays, facilitating quick, parallel evaluation of biological activity (MacBeath & Schreiber, 2000). This approach significantly minimizes the time and resources needed in the preliminary screening phase. Furthermore, nanocarriers augment the sensitivity and specificity of assays by elevating the local concentration of candidate drugs, thereby enhancing their interaction with target biomolecules (Rissin et al., 2010). Their nanoscale dimensions facilitate assay downsizing, conserve chemicals, and enable automation, thus expediting the screening procedure.

Nanocarriers markedly enhance the targeted delivery of drug candidates, hence improving the assessment of therapeutic efficacy and safety profiles during development. Encapsulating poorly soluble pharmaceuticals into nanocarriers enhances solubility and stability, thereby boosting absorption and bioavailability (Kumar et al., 2020). Nanocarriers can be functionalized with targeting ligands to direct medicine delivery to specific disease locations. This accuracy diminishes off-target effects and mitigates toxicity, facilitating more precise evaluations of a drug's therapeutic potential (Petros & DeSimone, 2010).

Nanocarriers can be designed for regulated or stimuli-responsive drug release, aiding researchers in the investigation of pharmacokinetics and pharmacodynamics under various situations. This control is crucial for enhancing medication release profiles and assessing therapeutic efficacy (Kamaly et al., 2016). Nanocarriers facilitate the simultaneous administration of various therapeutic drugs, hence streamlining the investigation of combination therapies. This feature is especially significant in oncology, where combination therapies are frequently required to improve therapeutic effectiveness (Alghamdi et al., 2022).

The integration of nanocarriers into pharmaceutical research enhances progress in personalized medicine. By customizing nanocarriers to target specific genetic or molecular markers, medicines can be designed that are more efficacious for individual patients. This method provides a more individualized treatment, enhancing both results and patient satisfaction (Wilhelm et al., 2016). Nanocarriers enhance the precision of drug behavior models in biological systems, hence diminishing attrition rates. They facilitate the early identification of ineffectiveness or toxicity, thereby averting late-stage failures and conserving essential time and resources (Moghimi et al., 2005).

Nanocarriers augment the therapeutic index of pharmaceuticals by increasing efficacy at the target location while reducing systemic exposure and adverse effects. This accuracy is essential for converting novel chemicals into effective clinical treatments (Peer et al., 2007). Nanocarriers can traverse biological barriers, including the BBB, which poses a considerable obstacle in administering medications to the central nervous system. Nanocarriers enable distribution to previously inaccessible locations, creating new opportunities for treating disorders impacting the brain and other regions shielded by barriers (Saraiva et al., 2016).

Nanocarriers' capacity to enhance therapeutic specificity and predictability may also streamline the regulatory approval process. Improved targeting accuracy and diminished toxicity profiles facilitate the demonstration of safety and efficacy to regulatory authorities, expediting the approval of novel medicines (Ventola, 2017).

1.5.3 Tissue Engineering and Regenerative Medicine

Nanocarriers have emerged as potent instruments in tissue engineering and regenerative medicine, facilitating the controlled and targeted delivery of growth factors, genes, and other bioactive compounds. Nanoscale systems improve tissue engineering effectiveness by safeguarding therapeutic compounds from degradation, facilitating controlled release, and influencing cellular activity crucial for tissue regeneration and repair. Growth factors are essential signaling molecules that govern activities including cell proliferation, differentiation, and migration, all of which are critical for tissue regeneration. Nanocarriers facilitate the regulated and prolonged release of growth factors at the designated spot, sustaining therapeutic concentrations for extended durations and minimizing the necessity for frequent administration (Briquez et al., 2015).

By encapsulating growth factors, nanocarriers safeguard these delicate molecules from enzymatic breakdown, preserving their biological activity. Furthermore, nanocarriers can be included in scaffolds or hydrogels, facilitating localized administration of growth factors. This focused strategy diminishes systemic exposure, hence decreasing side effects and enhancing therapeutic results (Langer & Peppas, 2003). Gene therapy has the capacity to induce cells to synthesize restorative proteins directly. Nanocarriers can transport genes that encode these proteins or growth factors to cells within tissue engineering constructions, facilitating prolonged synthesis of therapeutic proteins at the target location.

In contrast to viral vectors, which present safety concerns, nanocarriers offer a safer, non-viral alternative characterized by reduced immunogenicity and toxicity (Liu et al., 2011). Nanocarriers enhance cellular absorption and safeguard nucleic acids from degradation, therefore improving transfection efficiency and gene expression, which increases the efficacy of gene therapy in tissue engineering (Pack et al., 2005). Nanocarriers can be modified with bioactive ligands to enhance cell adhesion, proliferation, and differentiation. Modifying nanocarriers with RGD peptides (arginine–glycine–aspartic acid) improves cell

adhesion and integration into scaffolds (Hersel et al., 2003). Nanocarriers can transport chemicals that prompt stem cells to differentiate into specific cell types, thus aiding in the creation of functional tissue constructions (Silva et al., 2004).

Therapeutic proteins and nucleic acids are susceptible to enzymatic breakdown within the body. Nanocarriers provide a protective shield, guaranteeing that these bioactive chemicals arrive at their target cells unscathed. Polymeric nanocarriers, such as those composed of PLGA, offer a biodegradable framework that maintains the stability of medicinal medicines (Danhier et al., 2012). Lipid-based nanocarriers, including liposomes and solid LNPs, provide biocompatible environments that preserve the integrity of encapsulated molecules during transport (Plaza-Oliver et al., 2021).

Nanocarriers can be engineered to release therapeutic drugs in reaction to particular physiological cues, including pH, temperature, or enzymatic activity, commonly found in injured tissues. These intelligent delivery devices guarantee accurate release at the designated spot, enhancing efficacy while reducing side effects (Cheng et al., 2013). Moreover, stimuli-responsive systems can adjust to the healing process by releasing bioactive molecules corresponding to various stages of tissue repair (Uskoković & Ghosh, 2016).

The integration of nanocarriers into tissue engineering scaffolds promotes both the delivery of medicinal medicines and the characteristics of the scaffolds. Nanocarriers can transport angiogenic substances like vascular endothelial growth factors (VEGFs), facilitating the development of blood vessels essential for tissue viability and integration (Richardson et al., 2001). Nanocarriers can enhance the mechanical properties of scaffolds, hence reinforcing the structural integrity essential for efficient tissue regeneration (Shi et al., 2010).

Nanocarriers delivering bone morphogenetic proteins (BMPs) have demonstrated the ability to induce osteogenesis in bone defects, thus facilitating bone repair and mineralization (Zhao et al., 2010). This method has been effective in enhancing bone tissue engineering results. Chitosan-based nanocarriers containing transforming growth factor beta (TGF- β) have been employed to promote chondrogenic differentiation of stem cells, essential for cartilage tissue regeneration. This application demonstrates potential in cartilage restoration methodologies (Kim et al., 2023). Nanocarriers administering epidermal growth factor (EGF) augment keratinocyte proliferation and migration, expediting wound closure and facilitating skin regeneration. These carriers offer substantial advantages for wound healing and skin tissue engineering (Boateng & Catanzano, 2015).

1.5.4 Biosensors and Bioelectronics

Nanocarriers are essential for improving the sensitivity and efficacy of biosensors, enabling the detection of biomolecules at extremely low concentrations. Their unique characteristics, including elevated surface area, adjustable surface chemistries, and quantum phenomena, markedly enhance biosensor efficacy, rendering them exceptionally successful in diverse biomedical applications.

Gold and silver nanoparticles are frequently employed for their plasmonic properties, which increase optical signals in methods such as surface-enhanced Raman scattering (SERS) and surface plasmon resonance (SPR). These effects facilitate the identification of biomolecules at femtomolar concentrations, significantly enhancing sensitivity (Anker et al., 2008). Quantum dots, semiconductor nanoparticles with adjustable fluorescence, improve fluorescence-based detection owing to their luminosity and photostability. Conjugation of quantum dots to biomolecules enhances test sensitivity, enabling the detection of low-abundance targets (Medintz et al., 2005).

Nanocarriers provide an extensive surface area conducive to the immobilization of biorecognition components, including antibodies and aptamers. The extensive surface area enhances the probability of interaction with target molecules, thereby augmenting the sensitivity of biosensors (Li et al., 2019). Nanostructured materials, such as carbon nanotubes and graphene, are used in electrodes to augment electron transfer rates and enhance the sensitivity of electrochemical biosensors (Wang & Musameh, 2004).

In field-effect transistor (FET) biosensors, nanocarriers such as silicon nanowires or carbon nanotubes function as the conductive channel. Biomolecule binding to the surface alters the electrical characteristics of nanowires or nanotubes, facilitating label-free detection of biomolecules at extremely low concentrations (Patolsky et al., 2006).

Magnetic nanoparticles can be modified with capture ligands to extract and concentrate target proteins from intricate materials. This magnetic separation improves detection limits by concentrating the target molecules (Gijs, 2004). Magneto-resistive sensors can subsequently identify these magnetic nanoparticles, hence enhancing sensitivity (Rarokar et al., 2024).

Nanozymes—nanoscale carriers exhibiting enzyme-like activity—catalyze processes that produce observable signals, enhancing responses and facilitating ultra-sensitive detection (Wu et al., 2018). Nanocarriers can induce cascade reactions, resulting in many phases of signal amplification that enhance biosensor performance (Shi et al., 2016).

Nanocarriers are utilized in Förster resonance energy transfer (FRET) devices. Nanocarriers function as donors or acceptors in these systems, influencing energy transfer efficiency in the presence of target biomolecules, resulting in quantifiable fluorescence alterations and facilitating sensitive detection (Clapp et al., 2005). Nanocarriers can be integrated into lab-on-a-chip devices, improving the interaction between targets and detecting elements. These microfluidic devices minimize the necessary sample amount and provide swift, sensitive detection of biomolecules, proving particularly advantageous in point-of-care diagnostics (Sachdeva et al., 2021).

Gold nanoparticles facilitate electron transfer in glucose oxidase-based sensors, allowing for the detection of glucose at minimal concentrations. These sensors are very beneficial in diabetes care, offering accurate and timely glucose level measurement (Wu et al., 2007). DNA detection quantum dot-labeled probes, in conjunction with nanostructured electrodes, facilitate the identification of specific DNA sequences at picomolar concentrations. This methodology is essential for genetic diagnostics and the identification of gene mutations (Driskell et al., 2011).

Magnetic nanoparticles are utilized to isolate and concentrate low-abundance proteins from intricate biological materials, including blood. This improves the sensitivity of immunoassays for the early identification of illness biomarkers, facilitating earlier diagnosis of several disorders (Rarokar et al., 2024).

1.6 ADVANCES AND LIMITATIONS OF NANOCARRIERS

Understanding the benefits and constraints of nanocarriers is crucial for enhancing their design and utilization in therapeutic delivery, diagnostics, and biomedical research.

1.6.1 Improved Bioavailability and Targeting

Nanocarriers markedly enhance the solubility and stability of therapeutic medicines, particularly hydrophobic pharmaceuticals that face challenges due to inadequate water solubility. Encapsulating these medications within the hydrophobic core of nanocarriers, such as LNPs or polymeric micelles, enhances their dispersibility in biological fluids (Kalepu & Nekkanti, 2015). This encapsulation safeguards pharmaceuticals against enzymatic destruction and extends their circulation duration, enhancing their therapeutic efficacy. Nanocarriers augment the pharmacokinetic characteristics of pharmaceuticals by stabilizing them in circulation and enhancing their distribution. Surface changes, including PEGylation (the incorporation of PEG), diminish immune system recognition and extend circulation duration (Blanco et al., 2015). The prolonged circulation allows nanocarriers additional time to reach target tissues, hence improving therapeutic efficacy.

- *Passive Targeting*: This exploits the EPR effect, wherein tumors, characterized by their compromised vasculature and inadequate lymphatic drainage, facilitate the accumulation of nanocarriers (Maeda et al., 2008).
- *Active Targeting*: Nanocarriers can be functionalized with ligands, such as antibodies or peptides, that specifically bind to receptors overexpressed on target cells. This approach enhances cellular absorption by target cells and minimizes off-target effects, thereby augmenting therapy efficacy (Prajapati et al., 2024).

The dimensions and surface characteristics of nanocarriers facilitate cellular absorption, frequently via endocytosis. This is essential for administering medications that must function within cells, such as those aimed at cancer or viral infections (Oh & Park, 2009). Nanocarriers can surmount biological obstacles that restrict medicine delivery, including the BBB. Nanocarriers can traverse the BBB to administer medicines to the central nervous system by employing receptor-mediated transcytosis or by temporarily disrupting tight junctions between endothelial cells (Saraiva et al., 2016). Nanocarriers can be designed for controlled release, either progressively or in reaction to specific stimuli (e.g., pH, temperature, or enzyme presence). This guarantees that therapeutic concentrations are sustained at optimal levels for extended periods, hence decreasing dosage frequency and mitigating side effects (Kamaly et al., 2016).

Some nanocarriers, especially those composed of non-biodegradable materials, may induce chronic toxicity. Accumulation in organs, including the liver and kidneys, may result in inflammation or injury (Xuan et al., 2023). Comprehensive safety assessments are crucial to alleviating these hazards during the development phase. While PEGylation can diminish immune recognition, the recurrent application of nanocarriers may still elicit immunological responses, resulting in the generation of anti-drug antibodies (ADAs). These ADAs can neutralize the therapeutic drug, thus diminishing its efficacy, and may potentially induce hypersensitivity reactions (Talamini et al., 2021).

Attaining elevated drug loading and encapsulation efficiency continues to be a challenge, particularly with big or hydrophobic compounds. Insufficient drug-loading capacity may restrict treatment efficacy, necessitating more modification of nanocarrier designs to enhance delivery (Chehelgerdi et al., 2023). The

fabrication of nanocarriers can be costly and intricate, especially when employing sophisticated materials or surface alterations. Enhancing production capacity while preserving quality poses a hurdle for clinical translation. Furthermore, adhering to rigorous regulatory standards can impede the development and approval timeline (Al-Thani et al., 2024). Certain nanocarriers may exhibit instability under specific conditions, resulting in aggregation or premature payload release. This diminishes their shelf life and necessitates specific storage conditions, complicating general utilization (Su & Kang, 2020).

1.6.2 Controlled Release and Sustained Effect

Nanocarriers have emerged as fundamental components in contemporary treatments owing to their capacity to regulate drug release, sustain therapeutic concentrations over time, diminish the necessity for frequent dosing, and mitigate adverse effects. Enhancing treatment outcomes depends on the meticulous regulation of drug release kinetics, an essential element in various medicinal applications.

The design procedure of nanocarriers with tailored release kinetics commences with the selection of materials. Biodegradable polymers such as PLGA, polycaprolactone (PCL), and chitosan enable precise modulation of drug release through the alteration of degradation rates. The degradation rate of the polymer is influenced by factors including molecular weight, monomer makeup, and crystallinity, which subsequently impact the drug release profile, whether rapid or prolonged (Danhier et al., 2012). Lipid-derived materials, including solid LNPs and lipid nanocapsules, present an alternative solution. Modifying the lipid content and crystallinity of these carriers enables a regulated and steady release, especially advantageous for hydrophobic pharmaceuticals (Mehnert & Mäder, 2012).

Surface modification is equally significant in controlling drug release. Nanocarriers coated with hydrophilic polymers such as PEG attenuate drug transport, leading to prolonged therapeutic effects (Gref et al., 1994). Furthermore, the integration of stimuli-responsive polymers provides enhanced control. These polymers respond to environmental alterations, including pH, temperature, or enzymatic activity. pH-sensitive polymers release medications in acidic environments, such as tumors, facilitating targeted and regulated administration (Koo et al., 2011). The nanocarrier structure is a crucial element. Creating carriers with a core-shell structure, wherein the outer shell modulates drug diffusion and safeguards the inner core, facilitates meticulous control over the timing and location of drug release (Xu et al., 2015). Adjusting the porosity of nanocarriers, such as mesoporous silica nanoparticles, enables precise modulation of drug-loading capacity and release rates through alterations in pore size, volume, and surface area (Slowing et al., 2008).

The interplay between pharmaceuticals and polymers is essential for attaining the intended release profile. Pharmaceuticals can be covalently attached to the polymer matrix via degradable linkages, such as ester or amide bonds, facilitating a controlled release as these bonds progressively degrade (Cheng et al., 2023). Similarly, ionic interactions between the pharmaceutical agent and the carrier material, exemplified by polyelectrolyte complexes, facilitate the precise modulation of drug release rates through the alteration of contact strength (Li et al., 2001). Diverse encapsulation methods augment control over medication release. LbL construction, in which nanocarriers are fabricated by successively depositing oppositely charged polymers, facilitates meticulous regulation of the carrier's thickness and the timing of drug release (De Geest et al., 2008). Altering factors in emulsion-based synthesis techniques, such as solvent evaporation rates or emulsifier concentrations, can impact the size and drug distribution within nanocarriers, therefore affecting drug release (Bouchemal et al., 2004).

For intricate treatment approaches, nanocarriers can be designed to administer several pharmaceuticals. Sequential release systems are engineered to deliver medications at variable rates, typically by including layers within the carrier or employing materials that degrade at distinct velocities (Wang & Kohane, 2017). Co-delivery systems enhance this concept by facilitating the simultaneous or regulated release of many therapeutic agents, including medicines and genes, which is particularly advantageous for combination cancer therapy (Aborode et al., 2024). The release of drugs driven by external stimuli presents a more novel method of medication delivery. Nanocarriers containing magnetic nanoparticles can be remotely triggered to release their contents upon exposure to magnetic fields, allowing for precise control over the timing and location of medication delivery (Liu et al., 2019). Nanocarriers that react to ultrasound or light provide non-invasive and precisely regulated drug delivery (Kaushik et al., 2024).

The potential uses of nanocarriers are extensive. In cancer therapy, controlled release carriers facilitate the prolonged delivery of chemotherapeutic drugs directly to tumors, therefore reducing systemic toxicity while preserving effective therapeutic concentrations at the target location (Shi et al., 2016). Sustained drug release systems for chronic diseases, such as diabetes or hypertension, diminish the necessity for frequent dosage, hence enhancing patient compliance and treatment outcomes (Singh et al., 2019). Nanocarriers that release antigens in a regulated fashion can replicate the normal course of diseases, resulting in augmented immune responses and increased vaccine efficacy (Lozano et al., 2023).

Nonetheless, numerous obstacles persist. Increasing the production of nanocarriers while ensuring uniform release profiles is challenging due to inconsistencies in synthesis methods. Ensuring consistency across production batches is crucial for clinical application. Stability and storage provide considerable issues. Preserving the structural integrity of nanocarriers during storage while ensuring their release qualities necessitates meticulous regulation of environmental parameters such as temperature and humidity. Ultimately, compliance with regulatory criteria for nanocarrier-based therapeutics necessitates comprehensive safety and efficacy assessments, emphasizing constant quality control and reproducibility.

1.6.3 Scalability and Manufacturing Challenges

Increasing the production of nanocarriers while ensuring consistent quality is a significant problem as these technologies transition from laboratory research to clinical applications. With the rising demand for nanocarrier-based therapeutics, it is crucial to tackle the intricacies of large-scale manufacturing to guarantee their effective incorporation into clinical applications.

A primary problem in this transformation is attaining uniformity between batches. Fluctuations in essential factors, including size, shape, surface charge, and drug loading, can profoundly affect biological performance and therapeutic efficacy. Maintaining stringent control over these features during extensive production is challenging, frequently resulting in discrepancies that undermine the safety and efficacy of the final product (Ventola, 2017). A significant challenge is the scalability of manufacturing processes. Methods refined for small-scale laboratory manufacturing frequently do not directly apply to industrial sizes. Alterations in parameters such as mixing efficiency, heat transfer, and reaction kinetics during scale-up might modify the characteristics of nanocarriers, necessitating process redesign to maintain consistent product quality (Fernandes et al., 2023a).

Furthermore, the synthesis of nanocarriers entails intricate procedures such as nanoprecipitation, emulsification, and self-assembly, each necessitating meticulous regulation of various parameters. Consistently replicating these conditions on a wider scale poses considerable technological difficulties, frequently requiring specialized apparatus to sustain the precision attained in smaller batches (Mahajan & Bhattacharya, 2024). Quality control and characterization introduce an additional layer of complications. Although comprehensive characterization is essential for guaranteeing the safety and efficacy of nanocarriers, numerous analytical techniques employed in research environments are too time-consuming or expensive for industrial-scale manufacturing. Establishing scalable, swift, and dependable analytical procedures are crucial for preserving quality while ensuring efficiency (Gabor, 2011).

Regulatory compliance poses an additional barrier. Nanocarriers must adhere to rigorous regulatory standards that necessitate comprehensive documentation of production procedures, quality control protocols, and product specifications. Increasing output while complying with these criteria requires substantial expenditure in regulatory affairs and quality assurance (Anselmo & Mitragotri, 2019). Stability challenges, such as aggregation, degradation, and alterations in physicochemical properties with time, further hinder the large-scale manufacture of nanocarriers. Maintaining long-term stability during storage and distribution is essential for therapeutic applications, although it poses a considerable challenge that must be overcome to preserve product quality over time (Danaei et al., 2018).

The production expense becomes a significant challenge. The materials and techniques required for nanocarrier manufacture can be costly, and it is essential to balance cost with quality and efficacy to enhance the accessibility of nanocarrier-based therapeutics (Mitchell et al., 2020). Various solutions can be implemented to address these difficulties. Process optimization and standardization, via Quality by Design (QbD) and Good Manufacturing Practice (GMP), can delineate important parameters and institute consistent methods. Automation and sophisticated control systems can improve reproducibility and diminish human error (Ali et al., 2022).

Moreover, continuous manufacturing technologies, including microfluidics, present a viable option. Continuous flow techniques enhance control over the synthesis of nanocarriers, promoting scalable manufacturing and assuring uniformity. These approaches may also lower expenses relative to conventional batch production (Raval et al., 2019). Advanced analytical methods such as dynamic light scattering (DLS), nanoparticle tracking analysis (NTA), and high-performance liquid chromatography (HPLC) can be modified for high-throughput quality control, guaranteeing that goods comply with requisite requirements at industrial scales (Kaushik et al., 2023).

Enhancing the stability of nanocarriers via surface changes, lyophilization, and the incorporation of stabilizers is essential for maintaining long-term product quality. Excipients that inhibit degradation can further improve stability during storage (Kim et al., 2022). Engaging with regulatory agencies early in the development phase helps facilitate the pathway to market approval. Interacting with authorities guarantees that production procedures conform to regulatory standards, hence minimizing delays or impediments in the approval process (Anselmo & Mitragotri, 2019).

Strategies for cost reduction are of similar significance. Acquiring more economical raw materials, enhancing production methods, and increasing yields can reduce production costs, hence rendering nanocarrier-based therapies more cost-effective. As production increases, economies of scale are expected to decrease per-unit costs (Mitchell et al., 2020). Ultimately, investments in infrastructure and personnel training are essential for upholding superior manufacturing standards. Specialized facilities and proficient workers are crucial for maintaining consistent quality in large-scale nanocarrier production (Buya et al., 2024).

In summary, although the production of large-scale nanocarriers has certain problems, these can be mitigated through process optimization, new technologies, enhancements in quality control, regulatory coordination, and cost-effective solutions. With enough infrastructure and experience, it is feasible to scale production to satisfy clinical requirements.

1.7 REGULATORY CONSIDERATIONS

Understanding the regulatory framework is crucial for advancing nanocarriers from research to clinical use. Regulatory agencies are increasingly concentrating on the distinct characteristics of nanoparticles to guarantee the safety and efficacy of nanocarrier-based medicines. As nanomedicine progresses, regulatory frameworks pertaining to nanomaterials are adapting to confront the issues presented by these novel therapeutic systems.

1.7.1 Present Regulatory Framework for Nanomedicines

The regulatory framework for nanomedicines is influenced by the necessity to assess the unique properties of nanocarriers, encompassing their nanoscale dimensions, surface chemistry, and interactions with biological systems. These aspects affect the classification, assessment, and eventual approval of nanocarrier-based medicines.

Regulatory mandates profoundly influence the development and approval process for nanocarrier-based medicines, guaranteeing their safety, efficacy, and quality. These regulations affect multiple critical facets of growth. Regulatory agencies, such as the US FDA and the European Medicines Agency (EMA), have defined nanoparticles, influencing regulatory processes. The FDA characterizes nanomaterials as compounds with at least one dimension ranging from 1 to 100 nm (Food and Drug Administration, 2014). This classification imposes additional scrutiny and special regulatory guidelines on nanomedicines, affecting the evaluation of therapies for safety and efficacy.

Regulatory agencies necessitate comprehensive safety and efficacy assessments due to the distinctive characteristics of nanocarriers, including their augmented surface area, improved permeability, and potential for innovative biological interactions. Thorough preclinical investigations are essential to evaluate pharmacokinetics, biodistribution, toxicity, and interactions with biological systems (European Medicines Agency, 2011). Nanocarriers may pose unrecognized dangers, rendering comprehensive clinical testing essential. Regulatory bodies evaluate these risks using comprehensive clinical data to guarantee that nanocarriers do not present unexpected safety concerns.

Maintaining uniform quality in the production of nanocarriers is a significant regulatory issue. Agencies mandate compliance with GMP, ensuring stringent control over essential parameters influencing nanocarrier characteristics—such as size, surface charge, and drug encapsulation (Food and Drug Administration, 2017). Variability in these parameters may result in inconsistencies in efficacy and safety, rendering stringent quality control and manufacturing standards essential for clinical application.

Regulatory authorities, such as the FDA, have published guidance materials to elucidate regulatory expectations for developers. The FDA's advice on "Drug Products, Including Biological Products, that Contain Nanomaterials" delineates suggestions for the development, characterization, and assessment of nanomedicines (Food and Drug Administration, 2017). This guidance establishes a framework for evidencing safety, efficacy, and quality, influencing the entire development process from preclinical testing to clinical trials and quality assurance.

Nanocarrier-based therapies frequently exhibit attributes of both pharmaceuticals and devices, resulting in intricate regulatory processes. The dual nature of these goods can complicate the determination of their regulatory classification—whether as a medicine, biologic, or device. This complexity may prolong the evaluation process and necessitate further discussions (Bawa, 2011). Effectively traversing these routes necessitates a comprehensive comprehension of the various regulatory frameworks pertinent to the multifunctional characteristics of nanocarriers.

Developers aiming to launch nanocarrier-based medicines in worldwide marketplaces must adhere to diverse regulatory restrictions. Variations in regulatory mandates across areas, including the USA, Europe, and Asia, provide problems for achieving cross-border compliance (Tinkle et al., 2014). These variances might hinder the development process as companies must comply with the varied safety, efficacy, and quality criteria of each location. Attaining worldwide regulatory harmonization is a protracted goal; however, in the

interim, developers must customize their tactics to conform to the regulatory frameworks of each region to prevent delays in the approval process.

1.7.2 Safety Assessment of Nanocarriers

Evaluating the safety of nanocarriers necessitates customized techniques that account for their distinct physicochemical characteristics and possible long-term impacts on human health and the environment. Conventional toxicological approaches may inadequately address the intricacies of nanocarriers, requiring a thorough strategy to guarantee their safety and obtain regulatory approval. A thorough safety evaluation of nanocarriers must assess their interactions with biological systems, long-term effects, and environmental consequences. These evaluations must be customized to the unique characteristics of nanocarriers and encompass numerous essential elements.

The initial phase of safety assessment involves the comprehensive physicochemical characterization of nanocarriers. Essential aspects like size, shape, surface charge, aggregation state, and chemical composition must be meticulously analyzed, as these attributes affect the behavior of nanocarriers in biological systems (Guidance on Sample Preparation and Dosimetry for the Safety Testing of Manufactured Nanomaterials, 2012). For instance, size influences biodistribution and cellular absorption, whereas surface charge affects interactions with cell membranes and proteins. Comprehending these features is essential for forecasting the safety and effectiveness of nanocarriers.

In vitro toxicity studies serve as preliminary assessments to detect potential cytotoxicity, genotoxicity, and immunotoxicity. These investigations evaluate the interactions of nanocarriers with cells, proteins, and other biological molecules, providing preliminary markers of toxicity. Moreover, *in vitro* testing clarifies processes of cellular damage, including oxidative stress and membrane disruption, which can guide subsequent *in vivo* studies (Ceva et al., 2022). These tests are crucial for the early identification of risk prior to advancing to animal models.

Animal studies are essential for evaluating the biodistribution, pharmacokinetics, and toxicity of nanocarriers *in vivo*. Both acute and chronic toxicity should be assessed, emphasizing primary organs including the liver, kidneys, lungs, and brain (Fadell et al., 2018). These investigations offer insights into the metabolism, clearance, and possible accumulation of nanocarriers, which are essential for assessing their long-term safety in humans. The diminutive size of nanocarriers may result in tissue buildup over time, potentially causing long-term detrimental effects. Long-term exposure studies are essential to evaluate the persistence of nanocarriers in the body and the potential for delayed harm resulting from chronic accumulation (Oberdörster et al., 2005). These investigations are particularly significant for nanocarriers employed in chronic therapy or those capable of traversing biological barriers, such as the BBB.

Assessing the environmental impact of nanocarriers is essential, as they may infiltrate ecosystems during production, utilization, or disposal. Environmental assessments evaluate the degradability and ecological dangers of nanocarriers to confirm that they do not endanger the environment (Kah et al., 2019). These studies investigate the behavior of nanocarriers in aquatic, terrestrial, and atmospheric environments, along with their potential for bioaccumulation in species, while guaranteeing adherence to environmental safety regulations.

Nanocarriers can provoke immunological responses, rendering immunotoxicity studies a critical component of safety reviews. These evaluations encompass tests for complement activation, cytokine release, and hypersensitive reactions, aiding in the identification of unwanted immunological responses (Dobrovolskaia & McNeil, 2013). This testing is especially pertinent for nanocarriers that utilize materials or surface modifications potentially identified as foreign by the immune system.

Studies on reproductive and developmental toxicity are necessary to ascertain if nanocarriers adversely affect fertility or induce developmental anomalies. These studies are essential for nanocarriers employed in therapies for pregnant women or in prolonged treatments (Hardy et al., 2018). Assessing the potential for teratogenic effects and ensuring that nanocarriers do not disrupt reproductive processes are essential factors for regulatory approval.

To guarantee reproducibility and streamline regulatory approval, it is imperative to implement standardized and verified testing methodologies. International entities, such as the Organization for Economic Co-Operation and Development (OECD), have established protocols expressly for the assessment of nanomaterials (Guidance on Sample Preparation and Dosimetry for the Safety Testing of Manufactured Nanomaterials, 2012). These methodologies provide uniform safety assessments across diverse nanocarrier formulations, guaranteeing dependable and similar outcomes.

A thorough risk assessment methodology is essential for the safety evaluations of nanocarriers. This methodology integrates hazard identification, dose-response assessment, exposure evaluation, and risk characterization to deliver a comprehensive understanding of the possible dangers associated with

nanocarriers. Risk management techniques are formulated to mitigate recognized concerns, hence enabling the safe application of nanocarriers in clinical settings (Shatkin & Kim, 2015).

1.7.3 Ethical Considerations in Nanomedicine

Ethical considerations in nanomedicine, particularly regarding the development and use of nanocarrier technologies, are crucial for ensuring responsible and equitable progress. Critical concerns, including informed consent, equitable access, long-term societal ramifications, and regulatory monitoring, must be addressed to foster trust in nanomedicine and ensure its advantages are broadly disseminated across society. Comprehensive frameworks are essential to address the ethical concerns related to nanocarrier technologies, encompassing concepts such as patient autonomy, equity, beneficence, and justice. These frameworks ought to promote responsible innovation and societal involvement.

An ethical framework for nanomedicine must emphasize informed consent and maintain patient autonomy. Clear communication is crucial, as patients must be thoroughly educated about the attributes of nanocarrier-based therapy, encompassing the associated risks, advantages, and uncertainties (Lin & Allhoff, 2008). Due to the intricacies of nanotechnology, healthcare practitioners must guarantee that patients fully comprehend their therapy to make educated and independent decisions (Wasti et al., 2023).

Fair access to nanocarrier medicines is an essential element of any ethical framework. Access to nanomedicine may disproportionately advantage affluent individuals or nations, thereby intensifying healthcare disparities (Resnik & Tinkle, 2007). Ethical frameworks must facilitate equitable resource distribution, ensuring that marginalized populations are not overlooked. Internationally, these frameworks must also tackle access inequalities between rich and poor nations, promoting initiatives to guarantee global accessibility (Petersen et al., 2007).

Beneficence and nonmaleficence are essential ethical concepts in the utilization of nanocarrier technologies. Ethical frameworks must mandate thorough risk–benefit analyses to weigh potential advantages against hazards, ensuring that patient welfare is prioritized (Wasti et al., 2023). Furthermore, prolonged safety surveillance is crucial, as the impacts of nanocarriers may not be immediately apparent. Continuous supervision is required to safeguard patient health over time (Resnik & Tinkle, 2007).

Justice regarding nanocarrier technology entails guaranteeing equitable access and instituting stringent regulatory frameworks. Ethical frameworks must advocate for rigorous regulatory monitoring to protect public health (Mnyusiwalla et al., 2003). Moreover, societal engagement is vital. Engaging the public in debates about the consequences of nanomedicine cultivates trust and aligns scientific progress with societal values (Pidgeon & Rogers-Hayden, 2007).

The implementation of nanocarrier technologies may need the collection and analysis of sensitive patient information. Ethical frameworks must safeguard privacy and confidentiality, complying with data protection rules such as General Data Protection Regulation (GDPR) and Health Insurance Portability and Accountability Act (HIPAA) (Miller & Tucker, 2014). Patients should be certain that their health information is safeguarded and that nanocarriers do not infringe upon their privacy.

Ethical considerations pertain to the research and development of nanocarrier technology. Responsible innovation should direct research, ensuring transparency in reporting and a dedication to preventing conflicts of interest (Poel van de & Royakkers, 2011). Animal utilization in preclinical investigations must adhere to the 3Rs principles—Replacement, Reduction, and Refinement—to mitigate injury and guarantee ethical treatment (Ferdowsian & Beck, 2011). Ethical norms in research are fundamental to fostering trust and honesty in the development of nanomedicine.

Global cooperation is essential to align ethical norms and regulatory frameworks for nanomedicine. As nanocarrier technologies advance globally, ethical frameworks must encourage international collaboration to establish uniform regulatory processes, thereby facilitating the development of safe and effective nanomedicine worldwide (Hunt & Mehta, 2013).

1.8 FUTURE PERSPECTIVES

1.8.1 Emerging Trends in Nanocarrier Design

The domain of nanocarriers is advancing swiftly, with novel technologies poised to surmount existing constraints and transform therapeutic applications. Advancements in material science, biotechnology, and interdisciplinary collaboration are facilitating the creation of more advanced, efficient, and precise delivery systems. These developments are expected to enhance the efficacy, safety, and adaptability of nanocarriers in medicine, hence generating new opportunities for the treatment of complicated diseases.

Numerous advanced design methodologies demonstrate considerable potential in improving the accuracy, flexibility, and efficacy of nanocarriers for drug administration. Stimuli-responsive nanocarriers, referred to as “smart” nanocarriers, are designed to release therapeutic chemicals in reaction to particular internal

or external stimuli. These stimuli may encompass alterations in pH, redox gradients, enzymatic activity, temperature, light, or magnetic fields (Mura et al., 2013). In oncological therapy, pH-responsive systems have garnered interest owing to the acidic milieu characteristic of neoplastic tissues. These nanocarriers can selectively discharge their therapeutic payload at the tumor location, improving therapy efficacy while minimizing damage to healthy tissues (Sun et al., 2017). Enzyme-responsive nanocarriers utilize the overexpression of certain enzymes in pathological tissues to trigger medication release, thus enhancing targeted therapy (Hu et al., 2014).

Nanocarriers activated by external stimuli, such as light or magnetic fields, provide enhanced precision. These devices provide geographic and temporal manipulation of medication delivery through external controls, permitting non-invasive therapy of localized disorders. Light-activated carriers utilize photosensitive polymers to release pharmaceuticals upon exposure to light, rendering them appropriate for targeted therapy (Dai et al., 2013). Magnetic nanocarriers, containing magnetic nanoparticles, can be maneuvered and heated by magnetic fields to accurately deliver medications (Busquets et al., 2015).

Multifunctional nanocarriers are engineered to execute various functions—such as targeted distribution, diagnostics, and therapy—within a singular platform. These carriers are pivotal to theranostics, a discipline that integrates therapeutic and diagnostic activities (Kashyap et al., 2023). The capacity to incorporate real-time surveillance of drug administration and therapeutic outcomes is a significant benefit of multifunctional carriers, facilitating tailored therapies (Lim et al., 2014). Moreover, co-delivery systems that carry various therapeutic agents, including pharmaceuticals and gene treatments, can improve the treatment of complicated diseases such as cancer by leveraging synergistic effects (Hopkins et al., 2022).

Gold nanoparticles serve as multifunctional nanocarriers, functioning simultaneously as imaging agents and drug carriers, enhancing imaging quality and facilitating therapeutic delivery (Amaral et al., 2024). Furthermore, polymeric nanocarriers modified with targeting ligands can improve delivery specificity by attaching to cell receptors and transporting imaging chemicals for diagnostic applications (Bao et al., 2013). Biomimetic nanocarriers are engineered to replicate natural biological structures or processes, frequently by enveloping nanoparticles in cell membranes to evade immune recognition. These carriers utilize biological features to enhance therapeutic efficacy and safety (Fang et al., 2018).

Nanocarriers enveloped in red blood cell membranes can circumvent the immune system, prolonging their circulation duration and diminishing the probability of immunological reactions (Lotfabadi et al., 2024). Moreover, biomimetic carriers can acquire the targeting characteristics of the source cells, hence improving the specificity of medication delivery. Cancer cell membrane-coated nanoparticles can utilize homologous targeting to selectively return to and deliver medications to the original cancer cells (Chen et al., 2016). Instances of this biomimetic strategy encompass nanoparticles coated with red blood cell membranes, which merge the biocompatibility of red blood cells with the efficacy of synthetic nanocarriers, thus enhancing medication administration (Hu et al., 2011). Cancer cell membrane-coated nanoparticles utilize biological targeting mechanisms to improve therapy accuracy (Zhang et al., 2018).

Nanorobots exemplify an advanced methodology in the design of nanocarriers. These nanoscale devices possess the ability for autonomous movement and can traverse intricate biological environments to deliver treatments with unparalleled precision (Soto et al., 2020). Nanorobots possess the capability to transform targeted medicine delivery by accurately identifying sick cells and reducing off-target effects. These devices can be designed to react to certain biomarkers or environmental cues, administering medications alone at the designated locations. Externally guided active targeting technologies, such as magnetically driven nanorobots, enhance therapy precision by directing them to sick regions (Li et al., 2017). Enzyme-driven nanomotors are an innovation that uses biological enzymes for propulsion, enabling nanocarriers to traverse more efficiently toward designated targets. This method has potential for enhancing medication delivery in difficult settings, such as solid tumors (Xu et al., 2020).

1.8.2 Integration with Other Technologies

The amalgamation of nanocarriers with advanced technologies like artificial intelligence (AI) and 3D printing is set to transform their development and utilization in healthcare. Utilizing the analytical prowess of AI alongside the customization possibilities of 3D printing, researchers can markedly improve the design, optimization, and implementation of nanocarrier-based therapies, facilitating highly tailored and successful treatments.

The integration of AI and 3D printing with nanocarriers presents multiple benefits, fostering advancements in personalized medicine, precision manufacturing, and on-demand therapeutic solutions. AI enhances data-driven optimization in the design and development of nanocarriers. AI greatly improves the efficiency and accuracy of these systems using machine learning algorithms and predictive models. AI systems analyze extensive datasets concerning the characteristics and behaviors of nanocarriers. Employing machine learning models, AI can forecast interactions between nanocarriers and biological systems, facilitating the

advancement of more efficient and targeted medication delivery systems (Hamilton & Kingston, 2023). AI-driven predictive modeling enhances targeted efficiency by anticipating interactions with particular cell types, reducing side effects, and improving therapeutic precision (Tan et al., 2023).

AI can enhance HTS of nanocarrier formulations, swiftly finding the most suitable candidates for subsequent development (Serov & Vinogradov, 2022). This methodology decreases the time and expenses linked to conventional experimental techniques. Furthermore, AI can evaluate patient-specific data, customizing nanocarrier-based therapeutics for particular patients, hence promoting customized medicine (Kim et al., 2020). AI-driven methodologies enhance therapeutic outcomes by aligning the optimal nanocarrier system with a patient's genetic, physiological, and pathological characteristics.

3D printing facilitates the development of highly tailored nanocarrier-based drug delivery systems, providing enhanced control over design, composition, and drug release characteristics. This approach could transform the manufacture and utilization of nanocarriers in therapeutic environments. 3D printing offers unmatched accuracy in the fabrication of nanocarrier-loaded devices, enabling the creation of drug delivery systems with meticulously regulated geometry and compositions. This modification facilitates the creation of individualized dosage regimens and drug release profiles adapted to the specific requirements of each patient (Goole & Amighi, 2016). Furthermore, 3D printing facilitates the fabrication of intricate structures that integrate nanocarriers, applicable in tissue engineering and regenerative medicine (Murphy & Atala, 2014).

A primary advantage of 3D printing is its capacity to swiftly prototype nanocarrier systems, hence expediting research and development processes. This capacity enables researchers to rapidly iterate designs, optimizing formulations within a reduced timeframe (Lim et al., 2018). Furthermore, 3D printing facilitates point-of-care manufacturing, allowing for the on-site production of nanocarrier-based medicines, therefore mitigating storage and transit challenges and enhancing patient accessibility (Awad et al., 2019).

The amalgamation of AI and 3D printing cultivates a synergistic framework for enhancing the design, production, and utilization of nanocarriers. This integration facilitates adaptive, data-informed production processes that improve both accuracy and customization. AI-driven 3D printing facilitates real-time optimization of printing parameters and material choices, hence improving the efficacy of nanocarrier systems. Integrating AI algorithms into the 3D printing process enables researchers to forecast and modify variables like material qualities, layer thickness, and drug loading, so ensuring that the final product adheres to precise design parameters (Trenfield et al., 2018). This adaptive manufacturing method enhances consistency and quality, enabling the creation of high-performance nanocarriers for therapeutic applications. The integration of AI's data analysis capabilities with the customization potential of 3D printing enables the creation of patient-specific therapeutic solutions. Patient imaging data can be utilized to create 3D-printed implants or devices embedded with nanocarriers, customized to a patient's specific anatomical and clinical circumstances (Ozbolat & Yu, 2013). Moreover, the incorporation of sensors into these 3D-printed devices facilitates the development of intelligent, adaptive drug delivery systems. These devices can modulate drug release in real-time according to physiological feedback, providing a highly individualized and adaptive therapeutic strategy (Mandrycky et al., 2015).

1.8.3 Challenges and Opportunities in Clinical Translation

Connecting laboratory research with clinical practice poses hurdles and substantial prospects for nanocarrier technologies. The intricacy of these systems, regulatory challenges, biocompatibility issues, and financial impediments are significant constraints. Nonetheless, numerous ways can enhance their effective implementation in clinical environments. The therapeutic application of nanocarriers necessitates surmounting technical, regulatory, and economic obstacles. These challenges can be mitigated by interdisciplinary collaboration, regulatory involvement, sophisticated manufacturing techniques, and efficient stakeholder communication.

Nanocarriers exhibit considerable complexity; increasing their production while ensuring consistency presents substantial problems. The synthesis methods employed—namely, nanoprecipitation, emulsification, and self-assembly—render the attainment of batch-to-batch consistency challenging on an industrial scale (Fernandes et al., 2023b). Moreover, the characterization of their physicochemical features, such as size, shape, surface charge, and drug loading, is crucial for regulatory approval; however, it is a labor-intensive endeavor (Faria et al., 2018).

A significant impediment in clinical translation is the absence of defined regulatory frameworks for nanomedicine. The regulatory frameworks for nanocarriers are still developing, resulting in uncertainty in the approval process (Tinkle et al., 2014). Establishing long-term safety and efficacy is particularly difficult, as nanocarriers display distinct behaviors in biological systems that remain inadequately comprehended (Souto et al., 2024). This necessitates extensive preclinical and clinical investigations that adhere to rigorous safety and efficacy criteria.

Nanocarriers pose concerns including toxicity, immunogenicity, and off-target effects. These systems can elicit immunological responses, accumulate in tissues, or interact unpredictably with biological systems (Fadeel et al., 2018). Managing biodistribution and clearance presents a significant problem, as it is crucial to ensure that nanocarriers effectively reach their targets and are properly cleared from the body to ensure patient safety (Blanco et al., 2015).

Elevated development expenses represent a substantial impediment to the commercialization of nanocarrier technologies. The financial commitment necessary for research, development, and clinical trials may dissuade corporations from exploring these medicines (Etheridge et al., 2012). Moreover, it is essential to exhibit distinct advantages over current medicines to secure acceptability from healthcare professionals and patients, highlighting the necessity for cost-effectiveness and enhanced efficacy.

Interdisciplinary collaboration is fundamental for tackling the complexities of nanocarrier systems. Chemists, biologists, engineers, doctors, and regulatory specialists must collaborate to address technical, biological, and regulatory difficulties (Souto et al., 2024). Public–private collaborations can offer resources and enable the conversion of research into commercially viable goods. Establishing standardized techniques for the characterization and testing of nanocarriers can facilitate regulatory approval processes. This entails the formulation of explicit standards for safety, efficacy, and quality evaluations (Rauscher et al., 2019). Early and continuous engagement with regulatory bodies during the development phase ensures that nanocarrier development conforms to regulatory requirements, hence minimizing the possibility of approval delays (Anselmo & Mitragotri, 2019).

Utilizing sophisticated manufacturing technologies, including continuous manufacturing and microfluidic systems, can enhance the scalability and uniformity of nanoparticle carrier production (Shen et al., 2023). Continuous manufacturing enhances control over production processes, guaranteeing uniform product quality. Stringent quality control methods are essential to comply with regulatory criteria. Extensive safety evaluations, encompassing *in vitro* and *in vivo* research, are essential for the effective translation of nanocarriers. These investigations must assess toxicity, biodistribution, and immunological responses (Fadeel et al., 2018). Meticulously planned clinical trials are vital to establish the safety and efficacy of nanocarriers, which is requisite for regulatory approval and patient safety.

Engaging and instructing stakeholders—healthcare professionals, patients, and policymakers—is essential for promoting the acceptability of nanocarrier technology. Transparent communication about the advantages and disadvantages of nanocarriers fosters trust and facilitates informed decision-making (Etheridge et al., 2012). It is imperative to minimize the expenses related to nanocarrier development. Enhancing production processes to reduce waste and increase cost efficiency might decrease total development costs. Acquiring money via grants, investments, and governmental incentives can facilitate the advancement of nanocarrier-based therapeutics.

Notwithstanding the hurdles, substantial prospects exist for the clinical translation of nanocarrier technologies. One of the most potential uses is personalized medicine, wherein nanocarriers can be customized to administer medications according to unique patient profiles, hence enhancing therapeutic outcomes. Nanocarriers facilitate enhanced targeting and regulated drug release, hence minimizing negative effects and augmenting efficacy. Moreover, nanocarriers present novel therapeutic approaches, including gene therapy and immunotherapy, offering alternative solutions for conditions that are challenging to manage with traditional treatments.

1.9 CONCLUSIONS

Nanocarriers represent a significant advancement in the field of therapeutic nucleic acid and protein delivery. These systems offer highly controlled and targeted delivery mechanisms, overcoming many of the inherent challenges associated with traditional therapeutic approaches. Their ability to encapsulate and protect unstable molecules, such as mRNA, siRNA, and therapeutic proteins, while facilitating targeted delivery, underscores their transformative potential in contemporary medicine.

Nanocarriers enhance therapeutic efficacy through the optimization of several factors, including size, shape, surface charge, and functionalization. Such design parameters enable nanocarriers to penetrate biological barriers, evade immune system detection, and deliver therapeutic agents directly to target tissues. This minimizes off-target effects and improves bioavailability. Furthermore, advancements in nanotechnology have led to the development of stimuli-responsive nanocarrier systems, which release therapeutic agents in response to specific physiological conditions, thereby further enhancing treatment outcomes.

In the context of personalized medicine, nanocarriers hold the potential to revolutionize patient-specific therapies. By incorporating targeting ligands and customizable surface chemistries, these systems enable precise delivery of therapeutics based on an individual's molecular profile. This specificity has the potential

to reduce systemic toxicity and improve patient outcomes by tailoring treatments to the unique needs of each patient.

The clinical success of LNPs in mRNA vaccines, particularly during the COVID-19 pandemic, has demonstrated the viability of nanocarriers in achieving rapid global deployment and high efficacy. As drug delivery systems continue to be optimized and efforts to minimize side effects intensify, nanocarriers are poised to play a pivotal role in the future of biomedicine. Their scalability, safety, and therapeutic efficacy underline their importance in the advancement of gene and protein therapies, as well as their broader applications in areas such as tissue engineering and molecular diagnostics.

Continued research and innovation in the design of nanocarriers, particularly with regard to hybrid systems and biocompatible materials, are opening new opportunities for addressing previously untreatable diseases. As technological progress continues, the integration of nanocarriers with other advanced biomedical platforms may accelerate the development of therapies that are not only more effective but also less invasive, reshaping the future of healthcare.

The next frontier in nanocarrier technology will likely focus on improving biocompatibility, enhancing targeting specificity, and reducing potential immunogenicity. Advances in bioengineering and materials science will be essential to optimizing these systems for widespread clinical use. Future research will need to explore how nanocarrier technologies can be further refined to improve patient-specific therapies, lower treatment costs, and enhance the overall efficacy of medical interventions across various fields.

REFERENCES

- Aborode, A. T., Oluwajoba, A. S., Ibrahim, A. M., Ahmad, S., Mehta, A., Osayawe, O. J. et al. (2024). Nanomedicine in cancer therapy: advancing precision treatments. *Advances in Biomarker Sciences and Technology*, 6, 105–119. <https://doi.org/10.1016/j.abst.2024.06.003>
- Ahmad, A., Khan, J. M., & Haque, S. (2019). Strategies in the design of endosomolytic agents for facilitating endosomal escape in nanoparticles. *Biochimie*, 160, 61–75. <https://doi.org/10.1016/J.BIOCHI.2019.02.012>
- Alghamdi, M. A., Fallica, A. N., Virzì, N., Kesharwani, P., Pittalà, V., & Greish, K. (2022). The promise of nanotechnology in personalized medicine. *Journal of Personalized Medicine*, 12(5), 673. <https://doi.org/10.3390/jpm12050673>
- Ali, F., Neha, K., & Parveen, S. (2022). Current regulatory landscape of nanomaterials and nanomedicines: a global perspective. *Journal of Drug Delivery Science and Technology*, 80, 104118. <https://doi.org/10.1016/j.jddst.2022.104118>
- Alivisatos, P. (2003). The use of nanocrystals in biological detection. *Nature Biotechnology* 2004 22:1, 22(1), 47–52. <https://doi.org/10.1038/nbt927>
- Allen, T. M., & Cullis, P. R. (2013). Liposomal drug delivery systems: from concept to clinical applications. *Advanced Drug Delivery Reviews*, 65(1), 36–48. <https://doi.org/10.1016/J.ADDR.2012.09.037>
- Al-Thani, A. N., Jan, A. G., Abbas, M., Geetha, M., & Sadasivuni, K. K. (2024). Nanoparticles in cancer theragnostic and drug delivery: a comprehensive review. *Life Sciences*, 352, 122899. <https://doi.org/10.1016/j.lfs.2024.122899>
- Amaral, M. N., Nunes, D., Fortunato, E., Martins, R., Rodrigues, C., Faísca, P., et al. (2024). Gold nanoparticles for photothermal therapy – influence of experimental conditions on the properties of resulting AuNPs. *Journal of Drug Delivery Science and Technology*, 101, 106215. <https://doi.org/10.1016/j.jddst.2024.106215>
- Anitua, E., Alkhraisat, M. H., & Orive, G. (2011). Perspectives and challenges in regenerative medicine using plasma rich in growth factors. *Journal of Controlled Release*, 157(1), 29–38. <https://doi.org/10.1016/j.jconrel.2011.07.004>
- Anker, J. N., Hall, W. P., Lyandres, O., Shah, N. C., Zhao, J., & Van Duyne, R. P. (2008). Biosensing with plasmonic nanosensors. *Nature Materials*, 7(6), 442–453. <https://doi.org/10.1038/nmat2162>

- Anselmo, A. C., & Mitragotri, S. (2019). Nanoparticles in the clinic: an update. *Bioengineering & Translational Medicine*, 4(3), e10143. <https://doi.org/10.1002/btm2.10143>
- Awad, A., Fina, F., Trenfield, S., Patel, P., Goyanes, A., Gaisford, S., & Basit, A. (2019). 3D printed pellets (MiniPrintlets): A novel, multi-drug, controlled release platform technology. *Pharmaceutics*, 11(4), 148. <https://doi.org/10.3390/pharmaceutics11040148>
- Baden, L. R., El Sahly, H.M., Essink, B., Kotloff, K., Frey, S., Novak, R., et al. (2020). Efficacy and safety of the mRNA-1273 SARS-COV-2 vaccine. *New England Journal of Medicine*, 384(5), 403–416. <https://doi.org/10.1056/nejmoa2035389>
- Bahl, K., Senn, J. J., Yuzhakov, O., Bulychev, A., Brito, L. A., Hassett, K. J. et al. (2017). Preclinical and clinical demonstration of immunogenicity by mRNA vaccines against H10N8 and H7N9 influenza viruses. *Molecular Therapy*, 25(6), 1316–1327. <https://doi.org/10.1016/j.ymthe.2017.03.035>
- Bao, G., Mitragotri, S., & Tong, S. (2013). Multifunctional nanoparticles for drug delivery and molecular imaging. *Annual Review of Biomedical Engineering*, 15(1), 253–282. <https://doi.org/10.1146/annurev-bioeng-071812-152409>
- Bardhan, R., Lal, S., Joshi, A., & Halas, N. J. (2011). Theranostic nanoshells: from probe design to imaging and treatment of cancer. *Accounts of Chemical Research*, 44(10), 936–946. <https://doi.org/10.1021/AR200023X>
- Barenholz, Y. (2012). Doxil®--the first FDA-approved nano-drug: Lessons learned. *Journal of Controlled Release: Official Journal of the Controlled Release Society*, 160(2), 117–134. <https://doi.org/10.1016/J.JCONREL.2012.03.020>
- Bartel, D. P. (2004). MicroRNAs: genomics, biogenesis, mechanism, and function. *Cell*, 116(2), 281–297. [https://doi.org/10.1016/S0092-8674\(04\)00045-5](https://doi.org/10.1016/S0092-8674(04)00045-5)
- Bawa, R. (2011). Regulating nanomedicine – can the FDA handle it? *Current Drug Delivery*, 8(3), 227–234. <https://doi.org/10.2174/156720111795256156>
- Bechinger, B., & Lohner, K. (2006). Detergent-like actions of linear amphipathic cationic antimicrobial peptides. *Biochimica Et Biophysica Acta (BBA) – Biomembranes*, 1758(9), 1529–1539. <https://doi.org/10.1016/j.bbamem.2006.07.001>
- Behera, P. P., Kumar, N., Kumari, M., Kumar, S., Mondal, P. K., & Arun, R. K. (2023). Integrated microfluidic devices for point-of-care detection of bio-analytes and disease. *Sensors & Diagnostics*, 2(6), 1437–1459. <https://doi.org/10.1039/d3sd00170a>
- Benjaminsen, R. V., Mattheij, M. A., Henriksen, J. R., Moghimi, S. M., & Andresen, T. L. (2012). The possible “Proton sponge” effect of polyethylenimine (PEI) does not include change in lysosomal pH. *Molecular Therapy*, 21(1), 149–157. <https://doi.org/10.1038/mt.2012.185>
- Binauld, S., & Stenzel, M. H. (2013). Acid-degradable polymers for drug delivery: a decade of innovation. *Chemical Communications*, 49(21), 2082–2102. <https://doi.org/10.1039/C2CC36589H>
- Binnig, G., & Rohrer, H. (1984). Scanning tunneling microscopy. *Physica B+C*, 127(1–3), 37–45. [https://doi.org/10.1016/S0378-4363\(84\)80008-X](https://doi.org/10.1016/S0378-4363(84)80008-X)
- Blanco, E., Shen, H., & Ferrari, M. (2015). Principles of nanoparticle design for overcoming biological barriers to drug delivery. *Nature Biotechnology*, 33(9), 941–951. <https://doi.org/10.1038/nbt.3330>
- Boateng, J., & Catanzano, O. (2015). Advanced therapeutic dressings for effective wound healing—a review. *Journal of Pharmaceutical Sciences*, 104(11), 3653–3680. <https://doi.org/10.1002/jps.24610>

- Bouchemal, K., Briançon, S., Perrier, E., & Fessi, H. (2004). Nano-emulsion formulation using spontaneous emulsification: Solvent, oil and surfactant optimisation. *International Journal of Pharmaceutics*, 280(1–2), 241–251. <https://doi.org/10.1016/j.ijpharm.2004.05.016>
- Briquez, P. S., Hubbell, J. A., & Martino, M. M. (2015). Extracellular matrix-inspired growth factor delivery systems for skin wound healing. *Advances in Wound Care*, 4(8), 479–489. <https://doi.org/10.1089/wound.2014.0603>
- Busquets, M. A., Estelrich, J., & Sánchez-Martín, M. J. (2015). Nanoparticles in magnetic resonance imaging: from simple to dual contrast agents. *International Journal of Nanomedicine*, 10, 1727–1741. <https://doi.org/10.2147/ijn.s76501>
- Buya, A. B., Mahlangu, P., & Witika, B. A. (2024). From lab to industrial development of lipid nanocarriers using quality by design approach. *International Journal of Pharmaceutics X*, 8, 100266. <https://doi.org/10.1016/j.ijpx.2024.100266>
- Cevaál, P. M., Roche, M., Lewin, S. R., Caruso, F., & Faria, M. (2022). Experimental quantification of interactions between drug delivery systems and cells *in vitro*: a guide for preclinical nanomedicine evaluation. *Journal of Visualized Experiments*, 187. <https://doi.org/10.3791/64259>
- Chames, P., Van Regenmortel, M., Weiss, E., & Baty, D. (2009). Therapeutic antibodies: successes, limitations and hopes for the future. *British Journal of Pharmacology*, 157(2), 220–233. <https://doi.org/10.1111/j.1476-5381.2009.00190.x>
- Chehelgerdi, M., Chehelgerdi, M., Allela, O. Q. B., Pecho, R. D. C., Jayasankar, N., Rao, D. P. et al. (2023). Progressing nanotechnology to improve targeted cancer treatment: overcoming hurdles in its clinical implementation. *Molecular Cancer*, 22(1), 169. <https://doi.org/10.1186/S12943-023-01865-0>
- Chen, Z., Zhao, P., Luo, Z., Zheng, M., Tian, H., Gong, P., et al. (2016). Cancer cell membrane–biomimetic nanoparticles for homologous-targeting dual-modal imaging and photothermal therapy. *ACS Nano*, 10(11), 10049–10057. <https://doi.org/10.1021/acsnano.6b04695>
- Cheng, R., Meng, F., Deng, C., Klok, H., & Zhong, Z. (2013). Dual and multi-stimuli responsive polymeric nanoparticles for programmed site-specific drug delivery. *Biomaterials*, 34(14), 3647–3657. <https://doi.org/10.1016/j.biomaterials.2013.01.084>
- Cheng, R., Wang, S., & Santos, H. A. (2023). Acid-labile chemical bonds-based nanoparticles for endosome escape and intracellular delivery. *Biomedical Technology*, 3, 52–58. <https://doi.org/10.1016/j.bmt.2023.01.001>
- Cho, K., Wang, X., Nie, S., Chen, Z., & Shin, D. M. (2008). Therapeutic nanoparticles for drug delivery in cancer. *Clinical Cancer Research: An Official Journal of the American Association for Cancer Research*, 14(5), 1310–1316. <https://doi.org/10.1158/1078-0432.CCR-07-1441>
- Clapp, A. R., Medintz, I. L., & Mattoussi, H. (2005). Förster resonance energy transfer investigations using quantum-dot fluorophores. *ChemPhysChem*, 7(1), 47–57. <https://doi.org/10.1002/cphc.200500217>
- Cullis, P. R., & Hope, M. J. (2017). Lipid nanoparticle systems for enabling gene therapies. *Molecular Therapy*, 25(7), 1467–1475. <https://doi.org/10.1016/j.ymthe.2017.03.013>
- Dai, Y., Xiao, H., Liu, J., Yuan, Q., Ma, P., Yang, D. et al. (2013). In vivo multimodality imaging and cancer therapy by near-infrared light-triggered trans-platinum pro-drug-conjugated upconversion nanoparticles. *Journal of the American Chemical Society*, 135(50), 18920–18929. <https://doi.org/10.1021/ja410028q>
- Danaei, M., Dehghankhold, M., Ataei, S., Davarani, F. H., Javanmard, R., Dokhani, A. et al. (2018). Impact of particle size and polydispersity index on the clinical applications of lipidic nanocarrier systems. *Pharmaceutics*, 10(2), 57. <https://doi.org/10.3390/pharmaceutics10020057>

- Danhier, F., Ansorena, E., Silva, J. M., Coco, R., Breton, A. L., & Pr  at, V. (2012). PLGA-based nanoparticles: an overview of biomedical applications. *Journal of Controlled Release*, 161(2), 505–522. <https://doi.org/10.1016/j.jconrel.2012.01.043>
- De Geest, B. G., De Koker, S., Sukhorukov, G. B., Kreft, O., Parak, W. J., Skirtach, A. G. et al. (2008). Polyelectrolyte microcapsules for biomedical applications. *Soft Matter*, 5(2), 282–291. <https://doi.org/10.1039/b808262f>
- De Jong, S., Chikh, G., Sekirov, L., Raney, S., Semple, S., Klimuk, S. et al. (2007). Encapsulation in liposomal nanoparticles enhances the immunostimulatory, adjuvant and anti-tumor activity of subcutaneously administered CpG ODN. *Cancer Immunology, Immunotherapy: CII*, 56(8), 1251–1264. <https://doi.org/10.1007/S00262-006-0276-X>
- De Koker, S., Hoogenboom, R., & De Geest, B. G. (2012). Polymeric multilayer capsules for drug delivery. *Chemical Society Reviews*, 41(7), 2867. <https://doi.org/10.1039/c2cs15296g>
- Desnick, R. J., & Schuchman, E. H. (2002). Enzyme replacement and enhancement therapies: lessons from lysosomal disorders. *Nature Reviews Genetics*, 3(12), 954–966. <https://doi.org/10.1038/nrg963>
- Discher, D. E., & Eisenberg, A. (2002). Polymer vesicles. *Science*, 297(5583), 967–973. <https://doi.org/10.1126/SCIENCE.1074972>
- Dobrovolskaia, M. A., & McNeil, S. E. (2013). Understanding the correlation between in vitro and in vivo immunotoxicity tests for nanomedicines. *Journal of Controlled Release*, 172(2), 456–466. <https://doi.org/10.1016/j.jconrel.2013.05.025>
- Dreaden, E. C., Mac key, M. A., Huang, X., Kang, B., & El-Sayed, M. A. (2011). Beating cancer in multiple ways using nanogold. *Chemical Society Reviews*, 40(7), 3391–3404. <https://doi.org/10.1039/C0CS00180E>
- Driskell, J. D., Jones, C. A., Tompkins, S. M., & Tripp, R. A. (2011). One-step assay for detecting influenza virus using dynamic light scattering and gold nanoparticles. *The Analyst*, 136(15), 3083. <https://doi.org/10.1039/c1an15303j>
- Du, J., Lane, L. A., & Nie, S. (2015). Stimuli-responsive nanoparticles for targeting the tumor microenvironment. *Journal of Controlled Release*, 219, 205–214. <https://doi.org/10.1016/j.jconrel.2015.08.050>
- Duan, X., Xiao, J., Yin, Q., Zhang, Z., Yu, H., Mao, S., & Li, Y. (2013). Smart pH-sensitive and temporal-controlled polymeric micelles for effective combination therapy of doxorubicin and disulfiram. *ACS Nano*, 7(7), 5858–5869. <https://doi.org/10.1021/NN4010796>
- Duncan, R., & Gaspar, R. (2011). Nanomedicine(s) under the microscope. *Molecular Pharmaceutics*, 8(6), 2101–2141. <https://doi.org/10.1021/MP200394T>
- Ecker, D. M., Jones, S. D., & Levine, H. L. (2014). The therapeutic monoclonal antibody market. *MAbs*, 7(1), 9–14. <https://doi.org/10.4161/19420862.2015.989042>
- Elbashir, S. M., Harborth, J., Lendeckel, W., Yalcin, A., Weber, K., & Tuschl, T. (2001). Duplexes of 21-nucleotide RNAs mediate RNA interference in cultured mammalian cells. *Nature*, 411(6836), 494–498. <https://doi.org/10.1038/35078107>
- Elsababy, M., & Wooley, K. L. (2012). Design of polymeric nanoparticles for biomedical delivery applications. *Chemical Society Reviews*, 41(7), 2545. <https://doi.org/10.1039/c2cs15327k>
- Elumalai, K., Srinivasan, S., & Shanmugam, A. (2024). Review of the efficacy of nanoparticle-based drug delivery systems for cancer treatment. *Biomedical Technology*, 5, 109–122. <https://doi.org/10.1016/J.BMT.2023.09.001>

Etheridge, M. L., Campbell, S. A., Erdman, A. G., Haynes, C. L., Wolf, S. M., & McCullough, J. (2012). The big picture on nanomedicine: the state of investigational and approved nanomedicine products. *Nanomedicine Nanotechnology Biology and Medicine*, 9(1), 1–14. <https://doi.org/10.1016/j.nano.2012.05.013>

European Medicines Agency. (2011). *Reflection paper on nanotechnology-based medicinal products for human use* (Issue EMA/CHMP/79769/2011). https://etp-nanomedicine.eu/wp-content/uploads/2018/10/reflection-paper-nanotechnology-based-medicinal-products-human-use_en-1.pdf

Fadeel, B., Farcal, L., Hardy, B., Vázquez-Campos, S., Hristozov, D., Marcomini, A., et al. (2018). Advanced tools for the safety assessment of nanomaterials. *Nature Nanotechnology*, 13(7), 537–543. <https://doi.org/10.1038/s41565-018-0185-0>

Fan, D., Cao, Y., Cao, M., Wang, Y., Cao, Y., & Gong, T. (2023). Nanomedicine in cancer therapy. *Signal Transduction and Targeted Therapy*, 8(1), 1–12. <https://doi.org/10.1038/s41392-023-01536-y>

Fang, R. H., Kroll, A. V., Gao, W., & Zhang, L. (2018). Cell membrane coating nanotechnology. *Advanced Materials*, 30(23). <https://doi.org/10.1002/adma.201706759>

Faria, M., Björnmalm, M., Thurecht, K. J., Kent, S. J., Parton, R. G., Kavallaris, M., et al. (2018). Minimum information reporting in bio–nano experimental literature. *Nature Nanotechnology*, 13(9), 777–785. <https://doi.org/10.1038/s41565-018-0246-4>

Farkhani, S. M., Valizadeh, A., Karami, H., Mohammadi, S., Sohrabi, N., & Badrzadeh, F. (2014). Cell penetrating peptides: efficient vectors for delivery of nanoparticles, nanocarriers, therapeutic and diagnostic molecules. *Peptides*, 57, 78–94. <https://doi.org/10.1016/j.peptides.2014.04.015>

Farokhzad, O. C., & Langer, R. (2009). Impact of nanotechnology on drug delivery. *ACS Nano*, 3(1), 16–20. <https://doi.org/10.1021/NN900002M>

Fathi-Achachelouei, M., Knopf-Marques, H., Ribeiro da Silva, C. E., Barthès, J., Bat, E., Tezcaner, A. et al. (2019). Use of Nanoparticles in Tissue Engineering and Regenerative Medicine. *Frontiers in Bioengineering and Biotechnology*, 7, 453043. <https://doi.org/10.3389/FBIOE.2019.00113>

Ferdowsian, H. R., & Beck, N. (2011). Ethical and scientific considerations regarding animal testing and research. *PLOS One*, 6(9), e24059. <https://doi.org/10.1371/journal.pone.0024059>

Fernandes, C., Jathar, M., Sawant, B. K. S., & Warde, T. (2023a). Scale-up of nanoparticle manufacturing process. In *Pharmaceutical Process Engineering and Scale-up Principles* (pp. 173–203). Springer.

Fernandes, C., Jathar, M., Sawant, B. K. S., & Warde, T. (2023b). Scale-up of nanoparticle manufacturing process. In *AAPS Introductions in the Pharmaceutical Sciences* (pp. 173–203). https://doi.org/10.1007/978-3-031-31380-6_12

Feynman, R. P. (n.d.). *Plenty of Room at the Bottom*. Presentation at American Physical Society, Pasadena, December 1959.

Food and Drug Administration. (2014). *Guidance for industry considering whether an FDA-regulated product involves the application of nanotechnology contains nonbinding recommendations*. www.fda.gov/RegulatoryInformation/Guidances/ucm257698.htm

Food and Drug Administration. (2017). *Drug products, including biological products, that contain nanomaterials—guidance for industry*. www.fda.gov/regulatory-information/search-fda-guidance-documents/drug-products-including-biological-products-contain-nanomaterials-guidance-industry

Gabor, F. (2011). Characterization of nanoparticles intended for drug delivery. *Scientia Pharmaceutica*, 79(3), 701–702. <https://doi.org/10.3797/sciparm.br-11-01>

Gao, X., Cui, Y., Levenson, R. M., Chung, L. W. K., & Nie, S. (2004). In vivo cancer targeting and imaging with semiconductor quantum dots. *Nature Biotechnology*, 22(8), 969–976. <https://doi.org/10.1038/nbt994>

Geiss, G. K., Bumgarner, R. E., Birditt, B., Dahl, T., Dowidar, N., Dunaway, D. L. et al. (2008). Direct multiplexed measurement of gene expression with color-coded probe pairs. *Nature Biotechnology*, 26(3), 317–325. <https://doi.org/10.1038/nbt1385>

Geng, Y., Dalhaimer, P., Cai, S., Tsai, R., Tewari, M., Minko, T., & Discher, D. E. (2007). Shape effects of filaments versus spherical particles in flow and drug delivery. *Nature Nanotechnology* 2007 2:4, 2(4), 249–255. <https://doi.org/10.1038/nnano.2007.70>

Gijs, M. A. M. (2004). Magnetic bead handling on-chip: new opportunities for analytical applications. *Microfluid Nanofluid*, 1, 22–40. <https://doi.org/10.1007/s10404-004-0010-y>

Gilleron, J., Querbes, W., Zeigerer, A., Borodovsky, A., Marsico, G., Schubert, U. et al. (2013). Image-based analysis of lipid nanoparticle-mediated siRNA delivery, intracellular trafficking and endosomal escape. *Nature Biotechnology*, 31(7), 638–646. <https://doi.org/10.1038/NBT.2612>

Gisbert-Garzan, M., Lozano, D., Matsumoto, K., Komatsu, A., Manzano, M., Tamanoi, F., & Vallet-Regi, M. (2021). Designing mesoporous silica nanoparticles to overcome biological barriers by incorporating targeting and endosomal escape. *ACS Applied Materials and Interfaces*, 13(8), 9656–9666. Please replace with: <https://doi.org/10.1021/acsami.0c21507>

Goole, J., & Amighi, K. (2016). 3D printing in pharmaceuticals: a new tool for designing customized drug delivery systems. *International Journal of Pharmaceutics*, 499(1–2), 376–394. <https://doi.org/10.1016/j.ijpharm.2015.12.071>

Gratton, S. E. A., Ropp, P. A., Pohlhaus, P. D., Luft, J. C., Madden, V. J., Napier, M. E. et al. (2008). The effect of particle design on cellular internalization pathways. *Proceedings of the National Academy of Sciences of the United States of America*, 105(33), 11613–11618. <https://doi.org/10.1073/pnas.0801763105>

Gref, R., Minamitake, Y., Peracchia, M. T., Trubetskoy, V., Torchilin, V., & Langer, R. (1994). Biodegradable Long-Circulating polymeric nanospheres. *Science*, 263(5153), 1600–1603. <https://doi.org/10.1126/science.8128245>

Guo, S., Wang, J., Wang, Q., Wang, J., Qin, S., & Li, W. (2024). Advances in peptide-based drug delivery systems. *Heliyon*, 10(4), e26009. <https://doi.org/10.1016/J.HELİYON.2024.E26009>

Haddadzadegan, S., Dorkoosh, F., & Bernkop-Schnürch, A. (2022). Oral delivery of therapeutic peptides and proteins: technology landscape of lipid-based nanocarriers. *Advanced Drug Delivery Reviews*, 182, 114097. <https://doi.org/10.1016/j.addr.2021.114097>

Hadianamrei, R., & Zhao, X. (2022). Current state of the art in peptide-based gene delivery. *Journal of Controlled Release: Official Journal of the Controlled Release Society*, 343, 600–619. <https://doi.org/10.1016/J.JCONREL.2022.02.010>

Haghighat, Z. A., Safekordi, A., Ardjmand, M., & Akbarzadeh, A. (2024). Exploring the antitumor efficacy of PEGylated liposomes loaded with licorice extract for cancer therapy. *Current Cancer Drug Targets*, 24. <https://doi.org/10.2174/0115680096292153240416115744>

Hainfeld, J. F., Slatkin, D. N., & Smilowitz, H. M. (2004). The use of gold nanoparticles to enhance radiotherapy in mice. *Physics in Medicine and Biology*, 49(18), N309–N315. <https://doi.org/10.1088/0031-9155/49/18/n03>

Hamilton, S., & Kingston, B. R. (2023). Applying artificial intelligence and computational modeling to nanomedicine. *Current Opinion in Biotechnology*, 85, 103043. <https://doi.org/10.1016/j.copbio.2023.103043>

Hardy, A., Benford, D., Halldorsson, T., Jeger, M. J., Knutsen, H. K., More, S. et al. (2018). Guidance on risk assessment of the application of nanoscience and nanotechnologies in the food and feed chain: part 1, human and animal health. *EFSA Journal*, 16(7). <https://doi.org/10.2903/j.efsa.2018.5327>

- Harris, C. T., & Cohen, S. (2024). Reducing immunogenicity by design: approaches to minimize immunogenicity of monoclonal antibodies. *BioDrugs*, 38(2), 205–226. <https://doi.org/10.1007/s40259-023-00641-2>
- Hassett, K. J., Benenato, K. E., Jacquinet, E., Lee, A., Woods, A., Yuzhakov, O. et al. (2019). Optimization of lipid nanoparticles for intramuscular administration of mRNA vaccines. *Molecular Therapy—Nucleic Acids*, 15, 1–11. <https://doi.org/10.1016/j.omtn.2019.01.013>
- He, Q., Zhang, Z., Gao, F., Li, Y., & Shi, J. (2011). In vivo biodistribution and urinary excretion of mesoporous silica nanoparticles: effects of particle size and PEGylation. *Small (Weinheim an Der Bergstrasse, Germany)*, 7(2), 271–280. <https://doi.org/10.1002/SMLL.201001459>
- Hermeling, S., Crommelin, D. J. A., Schellekens, H., & Jiskoot, W. (2004). Structure-Immunogenicity relationships of therapeutic proteins. *Pharmaceutical Research*, 21(6), 897–903. <https://doi.org/10.1023/b:pham.0000029275.41323.a6>
- Hersel, U., Dahmen, C., & Kessler, H. (2003). RGD modified polymers: biomaterials for stimulated cell adhesion and beyond. *Biomaterials*, 24(24), 4385–4415. [https://doi.org/10.1016/s0142-9612\(03\)00343-0](https://doi.org/10.1016/s0142-9612(03)00343-0)
- Hopkins, C., Javius-Jones, K., Wang, Y., Hong, H., Hu, Q., & Hong, S. (2022). Combinations of chemo-, immuno-, and gene therapies using nanocarriers as a multifunctional drug platform. *Expert Opinion on Drug Delivery*, 19(10), 1337–1349. <https://doi.org/10.1080/17425247.2022.2112569>
- Horejs, C. (2021). From lipids to lipid nanoparticles to mRNA vaccines. *Nature Reviews. Materials*, 6(12), 1075–1076. <https://doi.org/10.1038/S41578-021-00379-9>
- Hou, X., Zaks, T., Langer, R., & Dong, Y. (2021). Lipid nanoparticles for mRNA delivery. *Nature Reviews Materials*, 6(12), 1078–1094. <https://doi.org/10.1038/s41578-021-00358-0>
- Hu, C. J., Zhang, L., Aryal, S., Cheung, C., Fang, R. H., & Zhang, L. (2011). Erythrocyte membrane-camouflaged polymeric nanoparticles as a biomimetic delivery platform. *Proceedings of the National Academy of Sciences*, 108(27), 10980–10985. <https://doi.org/10.1073/pnas.1106634108>
- Hu, Q., Katti, P. S., & Gu, Z. (2014). Enzyme-responsive nanomaterials for controlled drug delivery. *Nanoscale*, 6(21), 12273–12286. <https://doi.org/10.1039/C4NR04249B>
- Hunt, G., & Mehta, M. D. (Eds.). (2013). *Nanotechnology: Risk, Ethics and Law*. Routledge.
- Immordino, M. L., Dosio, F., & Cattel, L. (2006). Stealth liposomes: review of the basic science, rationale, and clinical applications, existing and potential. *International Journal of Nanomedicine*, 1(3), 297.
- Jackson, L. A., Anderson, E. J., Roupheal, N. G., Roberts, P. C., Makhene, M., Coler, R. N. et al. (2020). An mRNA vaccine against SARS-CoV-2—Preliminary report. *New England Journal of Medicine*, 383(20), 1920–1931. <https://doi.org/10.1056/nejmoa2022483>
- Jain, S., Hirst, D. G., & O’Sullivan, J. M. (2012). Gold nanoparticles as novel agents for cancer therapy. *The British Journal of Radiology*, 85(1010), 101. <https://doi.org/10.1259/BJR/59448833>
- Janib, S. M., Moses, A. S., & MacKay, J. A. (2010). Imaging and drug delivery using theranostic nanoparticles. *Advanced Drug Delivery Reviews*, 62(11), 1052–1063. <https://doi.org/10.1016/j.addr.2010.08.004>
- Jiang, W., Kim, B. Y. S., Rutka, J. T., & Chan, W. C. W. (2008). Nanoparticle-mediated cellular response is size-dependent. *Nature Nanotechnology*, 3(3), 145–150. <https://doi.org/10.1038/nnano.2008.30>
- Jokerst, J. V., Lobovkina, T., Zare, R. N., & Gambhir, S. S. (2011). Nanoparticle PEGylation for imaging and therapy. *Nanomedicine (London, England)*, 6(4), 715–728. <https://doi.org/10.2217/NNM.11.19>

- Kah, M., Tufenkji, N., & White, J. C. (2019). Nano-enabled strategies to enhance crop nutrition and protection. *Nature Nanotechnology*, 14(6), 532–540. <https://doi.org/10.1038/s41565-019-0439-5>
- Kalepu, S., & Nekkanti, V. (2015). Insoluble drug delivery strategies: review of recent advances and business prospects. *Acta Pharmaceutica Sinica B*, 5(5), 442–453. <https://doi.org/10.1016/j.apsb.2015.07.003>
- Kalluri, R., & LeBleu, V. S. (2020). The biology, function, and biomedical applications of exosomes. *Science (New York, N.Y.)*, 367(6478). <https://doi.org/10.1126/SCIENCE.AAU6977>
- Kamaly, N., Yameen, B., Wu, J., & Farokhzad, O. C. (2016). Degradable controlled-release polymers and polymeric nanoparticles: mechanisms of controlling drug release. *Chemical Reviews*, 116(4), 2602–2663. <https://doi.org/10.1021/acs.chemrev.5b00346>
- Kashyap, B. K., Singh, V. V., Solanki, M. K., Kumar, A., Ruokolainen, J., & Kesari, K. K. (2023). Smart nanomaterials in cancer theranostics: challenges and opportunities. *ACS Omega*, 8(16), 14290–14320. <https://doi.org/10.1021/acsomega.2c07840>
- Kataoka, K., Harada, A., & Nagasaki, Y. (2012). Block copolymer micelles for drug delivery: design, characterization and biological significance. *Advanced Drug Delivery Reviews*, 64(SUPPL.), 37–48. <https://doi.org/10.1016/J.ADDR.2012.09.013>
- Kaushik, A., Khan, S., Pharasi, N., & Mani, S. (2024). Dual pH and ultrasound responsive nanocarriers: a smart approach in cancer theranostics. *Journal of Drug Delivery Science and Technology*, 95, 105560. <https://doi.org/10.1016/j.jddst.2024.105560>
- Kaushik, A., Srinivasan, S. S., & Mishra, Y. K. (2023). *Analytical Techniques for Biomedical Nanotechnology*. IOP Publishing. <https://doi.org/10.1088/978-0-7503-3379-5>
- Kay, M. A., Glorioso, J. C., & Naldini, L. (2001). Viral vectors for gene therapy: the art of turning infectious agents into vehicles of therapeutics. *Nature Medicine*, 7(1), 33–40. <https://doi.org/10.1038/83324>
- Khan, I., Saeed, K., & Khan, I. (2019). Nanoparticles: properties, applications and toxicities. *Arabian Journal of Chemistry*, 12(7), 908–931. <https://doi.org/10.1016/J.ARABJC.2017.05.011>
- Khazaei, M., Hosseini, M. S., Haghighi, A. M., & Misaghi, M. (2023). Nanosensors and their applications in early diagnosis of cancer. *Sensing and Bio-Sensing Research*, 41, 100569. <https://doi.org/10.1016/J.SBSR.2023.100569>
- Khizar, S., Al-Dossary, A. A., Zine, N., Jaffrezic-Renault, N., Errachid, A., & Elaissari, A. (2022). Contribution of magnetic particles in molecular diagnosis of human viruses. *Talanta*, 241, 123243. <https://doi.org/10.1016/j.talanta.2022.123243>
- Kim, B., Hosn, R. R., Remba, T., Yun, D., Li, N., Abraham, W., Melo, M. B. et al. (2022). Optimization of storage conditions for lipid nanoparticle-formulated self-replicating RNA vaccines. *Journal of Controlled Release*, 353, 241–253. <https://doi.org/10.1016/j.jconrel.2022.11.022>
- Kim, H., Kim, E., Lee, I., Bae, B., Park, M., & Nam, H. (2020). Artificial intelligence in drug discovery: a comprehensive review of data-driven and machine learning approaches. *Biotechnology and Bioprocess Engineering*, 25(6), 895–930. <https://doi.org/10.1007/s12257-020-0049-y>
- Kim, S. C., Kim, D. W., Shim, Y. H., Bang, J. S., Oh, H. S., Kim, S. W. et al. (2001). In vivo evaluation of polymeric micellar paclitaxel formulation: toxicity and efficacy. *Journal of Controlled Release*, 72(1–3), 191–202. [https://doi.org/10.1016/s0168-3659\(01\)00275-9](https://doi.org/10.1016/s0168-3659(01)00275-9)
- Kim, Y., Zharkinbekov, Z., Raziyeva, K., Tabyldiyeva, L., Berikova, K., Zhumagul, D. et al. (2023). Chitosan-based biomaterials for tissue regeneration. *Pharmaceutics*, 15(3), 807. <https://doi.org/10.3390/pharmaceutics15030807>

Knauer, N., Pashkina, E., & Apartsin, E. (2019). Topological aspects of the design of nanocarriers for therapeutic peptides and proteins. *Pharmaceutics*, 11(2), 91. <https://doi.org/10.3390/pharmaceutics11020091>

Koo, H., Huh, M. S., Sun, I., Yuk, S. H., Choi, K., Kim, K., et al. (2011). In vivo targeted delivery of nanoparticles for theranosis. *Accounts of Chemical Research*, 44(10), 1018–1028. <https://doi.org/10.1021/ar2000138>

Kotzé, A. F., Lueßen, H. L., De Boer, A. G., Verhoef, J. C., & Junginger, H. E. (1999). Chitosan for enhanced intestinal permeability: prospects for derivatives soluble in neutral and basic environments. *European Journal of Pharmaceutical Sciences: Official Journal of the European Federation for Pharmaceutical Sciences*, 7(2), 145–151. [https://doi.org/10.1016/S0928-0987\(98\)00016-5](https://doi.org/10.1016/S0928-0987(98)00016-5)

Krishnan, U. M. (2020). *Protein and Peptide Nanostructures for Drug and Gene Delivery*. Elsevier. <https://doi.org/10.1016/b978-0-08-102850-6.00013-9>

Kumar, R., Dalvi, S. V., & Siril, P. F. (2020). Nanoparticle-based drugs and formulations: current status and emerging applications. *ACS Applied Nano Materials*, 3(6), 4944–4961. <https://doi.org/10.1021/acs.anm.0c00606>

Kwon, E. J., Bergen, J. M., & Pun, S. H. (2008). Application of an HIV GP41-derived peptide for enhanced intracellular trafficking of synthetic gene and siRNA delivery vehicles. *Bioconjugate Chemistry*, 19(4), 920–927. <https://doi.org/10.1021/bc700448h>

Langer, R., & Peppas, N. A. (2003). Advances in biomaterials, drug delivery, and bionanotechnology. *AIChE Journal*, 49(12), 2990–3006. <https://doi.org/10.1002/aic.690491202>

Lee, D., Koo, H., Sun, I., Ryu, J. H., Kim, K., & Kwon, I. C. (2011). Multifunctional nanoparticles for multimodal imaging and theragnosis. *Chemical Society Reviews*, 41(7), 2656–2672. <https://doi.org/10.1039/c2cs15261d>

Lee, E. S., Shin, H. J., Na, K., & Bae, Y. H. (2003). Poly(L-histidine)-PEG block copolymer micelles and pH-induced destabilization. *Journal of Controlled Release*, 90(3), 363–374. [https://doi.org/10.1016/S0168-3659\(03\)00205-0](https://doi.org/10.1016/S0168-3659(03)00205-0)

Lei, W., Yang, C., Wu, Y., Ru, G., He, X., Tong, X., et al. (2022). Nanocarriers surface engineered with cell membranes for cancer targeted chemotherapy. *Journal of Nanobiotechnology*, 20(1). <https://doi.org/10.1186/s12951-022-01251-w>

Li, J., De Ávila, B. E., Gao, W., Zhang, L., & Wang, J. (2017). Micro/nanorobots for biomedicine: delivery, surgery, sensing, and detoxification. *Science Robotics*, 2(4). <https://doi.org/10.1126/scirobotics.aam6431>

Li, S., Ma, L., Zhou, M., Li, Y., Xia, Y., Fan, X. et al. (2019). New opportunities for emerging 2D materials in bioelectronics and biosensors. *Current Opinion in Biomedical Engineering*, 13, 32–41. <https://doi.org/10.1016/j.cobme.2019.08.016>

Li, Y., & Champion, J. A. (2022). Self-assembling nanocarriers from engineered proteins: design, functionalization, and application for drug delivery. *Advanced Drug Delivery Reviews*, 189, 114462. <https://doi.org/10.1016/j.addr.2022.114462>

Li, Y., Pei, Y., Zhang, X., Gu, Z., Zhou, Z., Yuan, W., et al. (2001). PEGylated PLGA nanoparticles as protein carriers: synthesis, preparation and biodistribution in rats. *Journal of Controlled Release*, 71(2), 203–211. [https://doi.org/10.1016/s0168-3659\(01\)00218-8](https://doi.org/10.1016/s0168-3659(01)00218-8)

Lim, E., Kim, T., Paik, S., Haam, S., Huh, Y., & Lee, K. (2014). Nanomaterials for theranostics: recent advances and future challenges. *Chemical Reviews*, 115(1), 327–394. <https://doi.org/10.1021/cr300213b>

Lim, S. H., Kathuria, H., Tan, J. J. Y., & Kang, L. (2018). 3D printed drug delivery and testing systems — a passing fad or the future? *Advanced Drug Delivery Reviews*, 132, 139–168. <https://doi.org/10.1016/j.addr.2018.05.006>

- Lin, P., & Allhoff, F. (2008). *Nanotechnology & Society*. Springer. <https://doi.org/10.1007/978-1-4020-6209-4>
- Liu, B., Shi, T., Yang, H., Zhang, J., & Li, Y. (2024). Size-dependent effect on controllable release and field insecticidal efficacy of diamide insecticide polylactic acid microspheres delivery systems against *Ostrinia nubilalis*. *International Journal of Biological Macromolecules*, 278, 134659. <https://doi.org/10.1016/j.ijbio mac.2024.134659>
- Liu, J. F., Jang, B., Issadore, D., & Tsourkas, A. (2019). Use of magnetic fields and nanoparticles to trigger drug release and improve tumor targeting. *Wiley Interdisciplinary Reviews Nanomedicine and Nanobiotechnology*, 11(6). <https://doi.org/10.1002/wnan.1571>
- Liu, M., Wang, L., Lo, Y., Shiu, S. C. C., Kinghorn, A. B., & Tanner, J. A. (2022). Aptamer-enabled nanomaterials for therapeutics, drug targeting and imaging. *Cells*, 11(1). <https://doi.org/10.3390/CELLS11010159>
- Liu, X., Jin, X., & Ma, P. X. (2011). Nanofibrous hollow microspheres self-assembled from star-shaped polymers as injectable cell carriers for knee repair. *Nature Materials*, 10(5), 398–406. <https://doi.org/10.1038/nmat2999>
- Liu, Y., Miyoshi, H., & Nakamura, M. (2007). Nanomedicine for drug delivery and imaging: a promising avenue for cancer therapy and diagnosis using targeted functional nanoparticles. *International Journal of Cancer*, 120(12), 2527–2537. <https://doi.org/10.1002/ijc.22709>
- Longmire, M., Choyke, P. L., & Kobayashi, H. (2008). Clearance properties of nano-sized particles and molecules as imaging agents: considerations and caveats. *Nanomedicine*, 3(5), 703–717. <https://doi.org/10.2217/17435889.3.5.703>
- Lotfabad, A. S., Abadi, B., & Rezaei, N. (2024). Biomimetic nanotechnology for cancer immunotherapy: state of the art and future perspective. *International Journal of Pharmaceutics*, 123923. <https://doi.org/10.1016/j.ijpharm.2024.123923>
- Lou, D., Fan, L., Jiang, T., & Zhang, Y. (2022). Advances in nanoparticle-based lateral flow immunoassay for point-of-care testing. *View*, 3(1). <https://doi.org/10.1002/viw.20200125>
- Lozano, D., Larraga, V., Vallet-Regí, M., & Manzano, M. (2023). An overview of the use of nanoparticles in vaccine development. *Nanomaterials*, 13(12), 1828. <https://doi.org/10.3390/nano13121828>
- Lu, Y., & Low, P. S. (2012). Folate-mediated delivery of macromolecular anticancer therapeutic agents. *Advanced Drug Delivery Reviews*, 64(SUPPL.), 342–352. <https://doi.org/10.1016/J.ADDR.2012.09.020>
- MacBeath, G., & Schreiber, S. L. (2000). Printing proteins as microarrays for High-Throughput Function determination. *Science*, 289(5485), 1760–1763. <https://doi.org/10.1126/science.289.5485.1760>
- Maeda, H., Bharate, G., & Daruwalla, J. (2008). Polymeric drugs for efficient tumor-targeted drug delivery based on EPR-effect. *European Journal of Pharmaceutics and Biopharmaceutics*, 71(3), 409–419. <https://doi.org/10.1016/j.ejpb.2008.11.010>
- Maeda, H., Nakamura, H., & Fang, J. (2013). The EPR effect for macromolecular drug delivery to solid tumors: improvement of tumor uptake, lowering of systemic toxicity, and distinct tumor imaging in vivo. *Advanced Drug Delivery Reviews*, 65(1), 71–79. <https://doi.org/10.1016/J.ADDR.2012.10.002>
- Maeda, H., Wu, J., Sawa, T., Matsumura, Y., & Hori, K. (2000). Tumor vascular permeability and the EPR effect in macromolecular therapeutics: a review. *Journal of Controlled Release*, 65(1–2), 271–284. [https://doi.org/10.1016/S0168-3659\(99\)00248-5](https://doi.org/10.1016/S0168-3659(99)00248-5)
- Mahajan, K., & Bhattacharya, S. (2024). The advancement and obstacles in improving the stability of nanocarriers for precision drug delivery in the field of nanomedicine. *Current Topics in Medicinal Chemistry*, 24(8), 686–721. <https://doi.org/10.2174/0115680266287101240214071718>

- Mahler, H., Friess, W., Grauschopf, U., & Kiese, S. (2008). Protein aggregation: pathways, induction factors and analysis. *Journal of Pharmaceutical Sciences*, 98(9), 2909–2934. <https://doi.org/10.1002/jps.21566>
- Mandrycky, C., Wang, Z., Kim, K., & Kim, D. (2015). 3D bioprinting for engineering complex tissues. *Biotechnology Advances*, 34(4), 422–434. <https://doi.org/10.1016/j.biotechadv.2015.12.011>
- Mao, S., Sun, W., & Kissel, T. (2010). Chitosan-based formulations for delivery of DNA and siRNA. *Advanced Drug Delivery Reviews*, 62(1), 12–27. <https://doi.org/10.1016/J.ADDR.2009.08.004>
- Massignan, F., Gyulai, G., Horváti, K., Bősze, S., & Kiss, É. (2023). Effective nanoparticulate-type encapsulation delivery system for hydrophilic proteins and peptides. *EXPRESS Polymer Letters*, 18(1), 72–87. <https://doi.org/10.3144/expresspolymlett.2024.6>
- Medintz, I. L., Uyeda, H. T., Goldman, E. R., & Mattoussi, H. (2005). Quantum dot bioconjugates for imaging, labelling and sensing. *Nature Materials*, 4(6), 435–446. <https://doi.org/10.1038/nmat1390>
- Mehnert, W., & Mäder, K. (2012). Solid lipid nanoparticles. *Advanced Drug Delivery Reviews*, 64, 83–101. <https://doi.org/10.1016/j.addr.2012.09.021>
- Mehta, M., Bui, T. A., Yang, X., Aksoy, Y., Goldys, E. M., & Deng, W. (2023). Lipid-based nanoparticles for drug/gene delivery: an overview of the production techniques and difficulties encountered in their industrial development. *ACS Materials Au*, 3(6), 600–619. <https://doi.org/10.1021/acsmaterialsau.3c00032>
- Mendes, B. B., Connot, J., Avital, A., Yao, D., Jiang, X., Zhou, X. et al. (2022). Nanodelivery of nucleic acids. *Nature Reviews Methods Primers*, 2(1). <https://doi.org/10.1038/s43586-022-00104-y>
- Miao, Q., Xie, C., Zhen, X., Lyu, Y., Duan, H., Liu, X. et al. (2017). Molecular afterglow imaging with bright, biodegradable polymer nanoparticles. *Nature Biotechnology*, 35(11), 1102–1110. <https://doi.org/10.1038/nbt.3987>
- Miller, A. R., & Tucker, C. E. (2014). Privacy protection, personalized medicine and genetic testing. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.2411230>
- Mintzer, M. A., & Grinstaff, M. W. (2010). Biomedical applications of dendrimers: a tutorial. *Chemical Society Reviews*, 40(1), 173–190. <https://doi.org/10.1039/B901839P>
- Mitchell, M. J., Billingsley, M. M., Haley, R. M., Wechsler, M. E., Peppas, N. A., & Langer, R. (2020). Engineering precision nanoparticles for drug delivery. *Nature Reviews Drug Discovery*, 20(2), 101–124. <https://doi.org/10.1038/s41573-020-0090-8>
- Mizukami, S., Kashibe, M., Matsumoto, K., Hori, Y., & Kikuchi, K. (2017). Enzyme-triggered compound release using functionalized antimicrobial peptide derivatives. *Chemical Science*, 8(4), 3047–3053. <https://doi.org/10.1039/c6sc04435b>
- Mnyusiwalla, A., Daar, A. S., & Singer, P. A. (2003). Mind the gap: science and ethics in nanotechnology. *Nanotechnology*, 14(3), R9–R13. <https://doi.org/10.1088/0957-4484/14/3/201>
- Moghim, S. M., Hunter, A. C., & Murray, J. C. (2001). Long-circulating and target-specific nanoparticles: theory to practice. *Pharmacological Reviews*, 53(2), 283. <http://pharmrev.aspetjournals.org/content/53/2/283.abstract>
- Moghim, S. M., Hunter, A. C., & Murray, J. C. (2005). Nanomedicine: current status and future prospects. *FASEB Journal: Official Publication of the Federation of American Societies for Experimental Biology*, 19(3), 311–330. <https://doi.org/10.1096/FJ.04-2747REV>
- Mohan, A., Santhamoorthy, M., & Lee, Y. C. (2024). Recent advances in the pH-responsive organic–inorganic mesoporous hybrid silica for targeted drug delivery. *European Polymer Journal*, 206, 112783. <https://doi.org/10.1016/J.EURPOLYMJ.2024.112783>

- Mura, S., Nicolas, J., & Couvreur, P. (2013). Stimuli-responsive nanocarriers for drug delivery. *Nature Materials* 12:11, 12(11), 991–1003. <https://doi.org/10.1038/nmat3776>
- Murphy, S. V. & Atala, A. (2014). 3D bioprinting of tissues and organs. *Nature Biotechnology*, 32(8), 773–785. <https://doi.org/10.1038/nbt.2958>
- Muslykhah, U., Phupaboon, S., Suriyapha, C., Matra, M., & Wanapat, M. (2024). Encapsulation of protein-based bioactive from black soldier fly for ruminant feeding. *Journal of Agriculture and Food Research*, 18, 101325. <https://doi.org/10.1016/j.jafr.2024.101325>
- Nakase, I., Tanaka, G., & Futaki, S. (2012). Cell-penetrating peptides (CPPs) as a vector for the delivery of siRNAs into cells. *Molecular BioSystems*, 9(5), 855. <https://doi.org/10.1039/c2mb25467k>
- Nel, A., Xia, T., Mädler, L., & Li, N. (2006). Toxic potential of materials at the nanolevel. *Science (New York, NY)*, 311(5761), 622–627. <https://doi.org/10.1126/SCIENCE.1114397>
- Niculescu, A. G., & Grumezescu, A. M. (2022). Novel tumor-targeting nanoparticles for cancer treatment—a review. *International Journal of Molecular Sciences*, 23(9). <https://doi.org/10.3390/IJMS23095253>
- Nobs, L., Buchegger, F., Gurny, R., & Allémann, E. (2004). Current methods for attaching targeting ligands to liposomes and nanoparticles. *Journal of Pharmaceutical Sciences*, 93(8), 1980–1992. <https://doi.org/10.1002/JPS.20098>
- Oberdörster, G., Oberdörster, E., & Oberdörster, J. (2005). Nanotoxicology: an emerging discipline evolving from studies of ultrafine particles. *Environmental Health Perspectives*, 113(7), 823–839. <https://doi.org/10.1289/ehp.7339>
- OECD. (2012). *Guidance on sample preparation and dosimetry for the safety testing of manufactured nanomaterials*. <https://doi.org/10.1787/ed430e1d-en>
- Oh, Y. K., & Park, T. G. (2009). siRNA delivery systems for cancer treatment. *Advanced Drug Delivery Reviews*, 61(10), 850–862. <https://doi.org/10.1016/J.ADDR.2009.04.018>
- Olgenblum, G. I., Hutcheson, B. O., Pielak, G. J., & Harries, D. (2024). Protecting proteins from desiccation stress using molecular glasses and gels. *Chemical Reviews*, 124(9), 5668–5694. <https://doi.org/10.1021/acs.chemrev.3c00752>
- Ortiz, A. C., Jackson, N., Buelvas, N., Jerez, A., López-Muñoz, R. A., Morales, J. et al. (2024). Release kinetics approach of stimuli-responsive mesoporous silica nanocarriers: pH-sensitive linker versus pH-sensitive framework. *Journal of Drug Delivery Science and Technology*, 91, 105212. <https://doi.org/10.1016/j.jddst.2023.105212>
- Owens, D. E., & Peppas, N. A. (2006). Opsonization, biodistribution, and pharmacokinetics of polymeric nanoparticles. *International Journal of Pharmaceutics*, 307(1), 93–102. <https://doi.org/10.1016/J.IJPHARM.2005.10.010>
- Ozbolat, I. T., & Yu, N. Y. (2013). Bioprinting toward organ fabrication: challenges and future trends. *IEEE Transactions on Biomedical Engineering*, 60(3), 691–699. <https://doi.org/10.1109/tbme.2013.2243912>
- Pack, D. W., Hoffman, A. S., Pun, S., & Stayton, P. S. (2005). Design and development of polymers for gene delivery. *Nature Reviews Drug Discovery*, 4(7), 581–593. <https://doi.org/10.1038/nrd1775>
- Pandey, P., Arya, D. K., Deepak, P., Ali, D., Alarifi, S., Srivastava, S., et al. (2024). $\alpha\text{v}\beta 3$ integrin and folate-targeted pH-sensitive liposomes with dual ligand modification for metastatic breast cancer treatment. *Bioengineering*, 11(8), 800. <https://doi.org/10.3390/bioengineering11080800>
- Pardi, N., Hogan, M. J., Porter, F. W., & Weissman, D. (2018). mRNA vaccines—a new era in vaccinology. *Nature Reviews Drug Discovery*, 17(4), 261–279. <https://doi.org/10.1038/nrd.2017.243>

- Patel, D. M., Patel, N. N., & Patel, J. K. (2021). Nanomedicine scale-up technologies: feasibilities and challenges. In *Emerging Technologies for Nanoparticle Manufacturing* (pp. 511–539). Springer.
- Patolsky, F., Zheng, G., & Lieber, C. M. (2006). Nanowire-based biosensors. *Analytical Chemistry*, 78(13), 4260–4269. <https://doi.org/10.1021/ac069419j>
- Peer, D., Karp, J. M., Hong, S., Farokhzad, O. C., Margalit, R., & Langer, R. (2007). Nanocarriers as an emerging platform for cancer therapy. *Nature Nanotechnology* 2007 2:12, 2(12), 751–760. <https://doi.org/10.1038/nnano.2007.387>
- Petersen, A., Anderson, A., Wilkinson, C., & Allan, S. (2007). Nanotechnologies, risk and society. *Health Risk & Society*, 9(2), 117–124. <https://doi.org/10.1080/13698570701306765>
- Petros, R. A., & DeSimone, J. M. (2010). Strategies in the design of nanoparticles for therapeutic applications. *Nature Reviews Drug Discovery*, 9(8), 615–627. <https://doi.org/10.1038/nrd2591>
- Pidgeon, N., & Rogers-Hayden, T. (2007). Opening up nanotechnology dialogue with the publics: risk communication or “upstream engagement”? *Health Risk & Society*, 9(2), 191–210. <https://doi.org/10.1080/13698570701306906>
- Plaza-Oliver, M., Santander-Ortega, M. J., & Lozano, M. V. (2021). Current approaches in lipid-based nanocarriers for oral drug delivery. *Drug Delivery and Translational Research*, 11(2), 471–497. <https://doi.org/10.1007/s13346-021-00908-7>
- Poel van de, I. R., & Royakkers, L. M. M. (2011). *Ethics, Technology, and Engineering: An Introduction*. Wiley-Blackwell. <http://search.ebscohost.com/login.aspx?direct=true&scope=site&db=nlebk&db=nlabk&AN=510116>
- Polack, F. P., Thomas, S. J., Kitchin, N., Absalon, J., & others. (2020). Safety and efficacy of the BNT162B2 mRNA COVID-19 vaccine. *New England Journal of Medicine*, 383(27), 2603–2615. <https://doi.org/10.1056/nejmoa2034577>
- Prajapati, A., Rangra, S., Patil, R., Desai, N., Jyothi, V. G. S. S., Salave, S. et al. (2024). Receptor-Targeted nanomedicine for cancer therapy. *Receptors*, 3(3), 323–361. <https://doi.org/10.3390/receptors3030016>
- Presente, A., & Dowdy, S. F. (2013). PTD/CPP peptide-mediated delivery of siRNAs. *Current Pharmaceutical Design*, 19(16), 2943–2947. <https://doi.org/10.2174/1381612811319160008>
- Qin, X., Yu, C., Wei, J., Li, L., Zhang, C., Wu, Q., et al. (2019). Rational design of nanocarriers for intracellular protein delivery. *Advanced Materials*, 31(46). <https://doi.org/10.1002/adma.201902791>
- Rarokar, N., Yadav, S., Saoji, S., Bramhe, P., Agade, R., Gurav, S. et al. (2024). Magnetic nanosystem a tool for targeted delivery and diagnostic application: current challenges and recent advancement. *International Journal of Pharmaceutics X*, 7, 100231. <https://doi.org/10.1016/j.ijpx.2024.100231>
- Rauscher, H., Roebben, G., Mech, A., Gibson, P., Kestens, V., Linsinger, T. et al (2019). *An overview of concepts and terms used in the European Commission's definition of nanomaterial* (Issue EUR 29647 EN). <https://doi.org/10.2760/459136>
- Raval, N., Maheshwari, R., Kalyane, D., Youngren-Ortiz, S. R., Chougule, M. B., & Tekade, R. K. (2019). Importance of physicochemical characterization of nanoparticles in pharmaceutical product development. In *Elsevier eBooks* (pp. 369–400). <https://doi.org/10.1016/b978-0-12-817909-3.00010-8>
- Raza, K., Kumar, P., Kumar, N., & Malik, R. (2017). Pharmacokinetics and biodistribution of the nanoparticles. In Surendra Nimesh, Ramesh Chandra, and Nidhi Gupta (Eds.), *Advances in Nanomedicine for the Delivery of Therapeutic Nucleic Acids*, 165–186. Woodhead Publishing. <https://doi.org/10.1016/B978-0-08-100557-6.00009-2>

Rehman, U., Parveen, N., Sheikh, A., Abourehab, M. A. S., Sahebkar, A., & Kesharwani, P. (2022). Polymeric nanoparticles-siRNA as an emerging nano-polyplexes against ovarian cancer. *Colloids and Surfaces. B, Biointerfaces*, 218. <https://doi.org/10.1016/J.COLSURFB.2022.112766>

Resnik, D. B., & Tinkle, S. S. (2007). Ethics in nanomedicine. *Nanomedicine*, 2(3), 345–350. <https://doi.org/10.2217/17435889.2.3.345>

Richardson, T. P., Peters, M. C., Ennett, A. B., & Mooney, D. J. (2001). Polymeric system for dual growth factor delivery. *Nature Biotechnology*, 19(11), 1029–1034. <https://doi.org/10.1038/nbt1101-1029>

Rissin, D. M., Kan, C. W., Campbell, T. G., Howes, S. C., Fournier, D. R., Song, L. et al. (2010). Single-molecule enzyme-linked immunosorbent assay detects serum proteins at subfemtomolar concentrations. *Nature Biotechnology*, 28(6), 595–599. <https://doi.org/10.1038/nbt.1641>

Rojekar, S., Gholap, A. D., Togle, N., Bhoj, P., Haeck, C., Hatvate, N. et al. (2024). Current status of mannose receptor-targeted drug delivery for improved anti-HIV therapy. *Journal of Controlled Release*, 372, 494–521. <https://doi.org/10.1016/j.jconrel.2024.06.002>

Sachdeva, S., Davis, R. W., & Saha, A. K. (2021). Microfluidic Point-of-Care Testing: commercial landscape and future directions. *Frontiers in Bioengineering and Biotechnology*, 8. <https://doi.org/10.3389/fbioe.2020.602659>

Safari, J., & Zarnegar, Z. (2013). Advanced drug delivery systems: nanotechnology of health design A review. *Journal of Saudi Chemical Society*, 18(2), 85–99. <https://doi.org/10.1016/j.jscs.2012.12.009>

Saha, K., Agasti, S. S., Kim, C., Li, X., & Rotello, V. M. (2012). Gold nanoparticles in chemical and biological sensing. *Chemical Reviews*, 112(5), 2739–2779. <https://doi.org/10.1021/cr2001178>

Sahin, U., Karikó, K., & Türeci, Ö. (2014). mRNA-based therapeutics — developing a new class of drugs. *Nature Reviews Drug Discovery*, 13(10), 759–780. <https://doi.org/10.1038/nrd4278>

Salvati, A., Åberg, C., dos Santos, T., Varela, J., Pinto, P., Lynch, I. et al. (2011). Experimental and theoretical comparison of intracellular import of polymeric nanoparticles and small molecules: toward models of uptake kinetics. *Nanomedicine: Nanotechnology, Biology, and Medicine*, 7(6), 818–826. <https://doi.org/10.1016/J.NANO.2011.03.005>

Samaridou, E., Heyes, J., & Lutwyche, P. (2020). Lipid nanoparticles for nucleic acid delivery: current perspectives. *Advanced Drug Delivery Reviews*, 154–155, 37–63. <https://doi.org/10.1016/j.addr.2020.06.002>

Sanità, G., Carrese, B., & Lamberti, A. (2020). Nanoparticle surface functionalization: how to improve biocompatibility and cellular internalization. *Frontiers in Molecular Biosciences*, 7, 587012. <https://doi.org/10.3389/fmolb.2020.587012>

Saraiva, C., Praça, C., Ferreira, R., Santos, T., Ferreira, L., & Bernardino, L. (2016). Nanoparticle-mediated brain drug delivery: overcoming blood-brain barrier to treat neurodegenerative diseases. *Journal of Controlled Release: Official Journal of the Controlled Release Society*, 235, 34–47. <https://doi.org/10.1016/J.JCONREL.2016.05.044>

Serov, N., & Vinogradov, V. (2022). Artificial intelligence to bring nanomedicine to life. *Advanced Drug Delivery Reviews*, 184, 114194. <https://doi.org/10.1016/j.addr.2022.114194>

Shatkin, J. A., & Kim, B. (2015). Cellulose nanomaterials: life cycle risk assessment, and environmental health and safety roadmap. *Environmental Science Nano*, 2(5), 477–499. <https://doi.org/10.1039/c5en00059a>

Shen, Y., Gwak, H., & Han, B. (2023). Advanced manufacturing of nanoparticle formulations of drugs and biologics using microfluidics. *The Analyst*, 149(3), 614–637. <https://doi.org/10.1039/d3an01739g>

- Shi, J., Kantoff, P. W., Wooster, R., & Farokhzad, O. C. (2016). Cancer nanomedicine: progress, challenges and opportunities. *Nature Reviews. Cancer*, 17(1), 20–37. <https://doi.org/10.1038/nrc.2016.108>
- Shi, K., Dou, B., Yang, J., Yuan, R., & Xiang, Y. (2016). Cascaded strand displacement for non-enzymatic target recycling amplification and label-free electronic detection of microRNA from tumor cells. *Analytica Chimica Acta*, 916, 1–7. <https://doi.org/10.1016/j.aca.2016.02.034>
- Shi, L., Zhang, J., Zhao, M., Tang, S., Cheng, X., Zhang, W. et al. (2021). Effects of polyethylene glycol on the surface of nanoparticles for targeted drug delivery. *Nanoscale*, 13(24), 10748–10764. <https://doi.org/10.1039/D1NR02065J>
- Shi, M., Kretlow, J. D., Nguyen, A., Young, S., Baggett, L. S., Wong, M. E., et al. (2010). Antibiotic-releasing porous polymethylmethacrylate constructs for osseous space maintenance and infection control. *Biomaterials*, 31(14), 4146–4156. <https://doi.org/10.1016/j.biomaterials.2010.01.112>
- Shi, P., Cheng, Z., Zhao, K., Chen, Y., Zhang, A., Gan, W., & Zhang, Y. (2023). Active targeting schemes for nano-drug delivery systems in osteosarcoma therapeutics. *Journal of Nanobiotechnology*, 21(1), 1–27. <https://doi.org/10.1186/S12951-023-01826-1>
- Shire, S. J., Shahrokh, Z., & Liu, J. (2004). Challenges in the development of high protein concentration formulations. *Journal of Pharmaceutical Sciences*, 93(6), 1390–1402. <https://doi.org/10.1002/jps.20079>
- Shirian, J., Hockla, A., Gleba, J. J., Coban, M., Rotenberg, N., Strik, L. M., et al. (2024). Improving circulation Half-Life of therapeutic candidate N-TIMP2 by unfolded peptide extension. *Biomolecules*, 14(9), 1187. <https://doi.org/10.3390/biom14091187>
- Shubayev, V. I., Pisanic, T. R., & Jin, S. (2009). Magnetic nanoparticles for theragnostics. *Advanced Drug Delivery Reviews*, 61(6), 467–477. <https://doi.org/10.1016/j.addr.2009.03.007>
- Silva, G. A., Czeisler, C., Niece, K. L., Beniash, E., Harrington, D. A., Kessler, J. A. et al. (2004). Selective differentiation of neural progenitor cells by High-Epitope Density nanofibers. *Science*, 303(5662), 1352–1355. <https://doi.org/10.1126/science.1093783>
- Singh, A. P., Biswas, A., Shukla, A., & Maiti, P. (2019). Targeted therapy in chronic diseases using nanomaterial-based drug delivery vehicles. *Signal Transduction and Targeted Therapy*, 4(1). <https://doi.org/10.1038/s41392-019-0068-3>
- Skowicki, M., Tarvirdipour, S., Kraus, M., Schoenenberger, C. A., & Palivan, C. G. (2024). Nanoassemblies designed for efficient nuclear targeting. *Advanced Drug Delivery Reviews*, 211, 115354. <https://doi.org/10.1016/J.ADDR.2024.115354>
- Slowing, I. I., Vivero-Escoto, J. L., Wu, C., & Lin, V. S. (2008). Mesoporous silica nanoparticles as controlled release drug delivery and gene transfection carriers. *Advanced Drug Delivery Reviews*, 60(11), 1278–1288. <https://doi.org/10.1016/j.addr.2008.03.012>
- Soe, T. H., Watanabe, K., & Ohtsuki, T. (2020). Photoinduced endosomal escape mechanism: a view from photochemical internalization mediated by CPP-photosensitizer conjugates. *Molecules*, 26(1), 36. <https://doi.org/10.3390/molecules26010036>
- Soto, F., Wang, J., Ahmed, R., & Demirci, U. (2020). Medical micro/nanorobots in precision medicine. *Advanced Science*, 7(21). <https://doi.org/10.1002/advs.202002203>
- Souto, E. B., Blanco-Llamero, C., Krambeck, K., Kiran, N. S., Yashaswini, C., Postwala, H. et al. (2024). Regulatory insights into nanomedicine and gene vaccine innovation: safety assessment, challenges, and regulatory perspectives. *Acta Biomaterialia*, 180, 1–17. <https://doi.org/10.1016/j.actbio.2024.04.010>
- Su, S., & Kang, P. M. (2020). Recent advances in nanocarrier-assisted therapeutics delivery systems. *Pharmaceutics*, 12(9), 837. <https://doi.org/10.3390/pharmaceutics12090837>

- Sun, D., & Lu, Z. (2023). Structure and function of cationic and ionizable lipids for nucleic acid delivery. *Pharmaceutical Research*, 40(1), 27–46. <https://doi.org/10.1007/s11095-022-03460-2>
- Sun, Q., Ojha, T., Kiessling, F., Lammers, T., & Shi, Y. (2017). Enhancing tumor penetration of nanomedicines. *Biomacromolecules*, 18(5), 1449–1459. <https://doi.org/10.1021/acs.biomac.7b00068>
- Talamini, L., Matsuura, E., De Cola, L., & Muller, S. (2021). Immunologically inert nanostructures as selective therapeutic tools in inflammatory diseases. *Cells*, 10(3), 707. <https://doi.org/10.3390/cells10030707>
- Tan, P., Chen, X., Zhang, H., Wei, Q., & Luo, K. (2023). Artificial intelligence aids in development of nanomedicines for cancer management. *Seminars in Cancer Biology*, 89, 61–75. <https://doi.org/10.1016/j.semcancer.2023.01.005>
- Tchessalov, S., Maglio, V., Kazarin, P., Alexeenko, A., Bhatnagar, B., Sahni, E., et al. (2023). Practical advice on scientific design of freeze-drying process: 2023 update. *Pharmaceutical Research*, 40(10), 2433–2455. <https://doi.org/10.1007/s11095-023-03607-9>
- Tinkle, S., McNeil, S. E., Mühlebach, S., Bawa, R., Borchard, G., Barenholz, Y. et al. (2014). Nanomedicines: addressing the scientific and regulatory gap. *Annals of the New York Academy of Sciences*, 1313(1), 35–56. <https://doi.org/10.1111/nyas.12403>
- Todaro, B., Ottalagana, E., Luin, S., & Santi, M. (2023). Targeting peptides: the new generation of targeted drug delivery systems. *Pharmaceutics*, 15(6), 1648. <https://doi.org/10.3390/pharmaceutics15061648>
- Torchilin, V. P. (2014). Multifunctional, stimuli-sensitive nanoparticulate systems for drug delivery. *Nature Reviews Drug Discovery*, 13(11), 813–827. <https://doi.org/10.1038/nrd4333>
- Toy, R., Peiris, P. M., Ghaghada, K. B., & Karathanasis, E. (2014). Shaping cancer nanomedicine: the effect of particle shape on the in vivo journey of nanoparticles. *Nanomedicine (London, England)*, 9(1), 121. <https://doi.org/10.2217/NNM.13.191>
- Trenfield, S. J., Awad, A., Goyanes, A., Gaisford, S., & Basit, A. W. (2018). 3D printing pharmaceuticals: drug development to frontline care. *Trends in Pharmacological Sciences*, 39(5), 440–451. <https://doi.org/10.1016/j.tips.2018.02.006>
- Uskoković, V., & Ghosh, S. (2016). Carriers for the tunable release of therapeutics: etymological classification and examples. *Expert Opinion on Drug Delivery*, 13(12), 1729–1741. <https://doi.org/10.1080/17425247.2016.1200558>
- Varkouhi, A. K., Scholte, M., Storm, G., & Haisma, H. J. (2010). Endosomal escape pathways for delivery of biologicals. *Journal of Controlled Release*, 151(3), 220–228. <https://doi.org/10.1016/j.jconrel.2010.11.004>
- Ventola, C. L. (2017). Progress in nanomedicine: approved and investigational nanodrugs. *P&T: A Peer-Reviewed Journal for Formulary Management*, 42(12), 742–755. <https://pubmed.ncbi.nlm.nih.gov/29234213/>
- Veronese, F. M., & Pasut, G. (2005). PEGylation, successful approach to drug delivery. *Drug Discovery Today*, 10(21), 1451–1458. [https://doi.org/10.1016/s1359-6446\(05\)03575-0](https://doi.org/10.1016/s1359-6446(05)03575-0)
- Wang, J., & Musameh, M. (2004). Carbon nanotube screen-printed electrochemical sensors. *The Analyst*, 129(1), 1. <https://doi.org/10.1039/b313431h>
- Wang, W., Singh, S., Zeng, D. L., King, K., & Nema, S. (2006). Antibody structure, instability, and formulation. *Journal of Pharmaceutical Sciences*, 96(1), 1–26. <https://doi.org/10.1002/jps.20727>
- Wang, Y., & Kohane, D. S. (2017). External triggering and triggered targeting strategies for drug delivery. *Nature Reviews Materials*, 2(6). <https://doi.org/10.1038/natrevmats.2017.20>
- Wasti, S., Lee, I. H., Kim, S., Lee, J., & Kim, H. (2023). Ethical and legal challenges in nanomedical innovations: a scoping review. *Frontiers in Genetics*, 14. <https://doi.org/10.3389/fgene.2023.1163392>

- Wilhelm, S., Tavares, A. J., Dai, Q., Ohta, S., Audet, J., Dvorak, H. F. et al (2016). Analysis of nanoparticle delivery to tumours. *Nature Reviews Materials*, 1(5). <https://doi.org/10.1038/natrevmats.2016.14>
- Won, Y. W., Yoon, S. M., Lee, K. M., & Kim, Y. H. (2011). Poly(oligo-D-arginine) with internal disulfide linkages as a cytoplasm-sensitive carrier for siRNA delivery. *Molecular Therapy: The Journal of the American Society of Gene Therapy*, 19(2), 372–380. <https://doi.org/10.1038/MT.2010.242>
- Wu, B., Hou, S., Yin, F., Zhao, Z., Wang, Y., Wang, X., et al. (2007). Amperometric glucose biosensor based on multilayer films via layer-by-layer self-assembly of multi-wall carbon nanotubes, gold nanoparticles and glucose oxidase on the Pt electrode. *Biosensors and Bioelectronics*, 22(12), 2854–2860. <https://doi.org/10.1016/j.bios.2006.11.028>
- Wu, J., Wang, X., Wang, Q., Lou, Z., Li, S., Zhu, Y. et al. (2018). Nanomaterials with enzyme-like characteristics (nanozymes): next-generation artificial enzymes (II). *Chemical Society Reviews*, 48(4), 1004–1076. <https://doi.org/10.1039/c8cs00457a>
- Xu, H., Medina-Sánchez, M., Maitz, M. F., Werner, C., & Schmidt, O. G. (2020). Sperm micromotors for cargo delivery through flowing blood. *ACS Nano*, 14(3), 2982–2993. <https://doi.org/10.1021/acsnano.9b07851>
- Xu, Q., Xiao, Z., Yang, Q., Yu, T., Deng, X., Chen, N. et al (2024). Hydrogel-based cardiac repair and regeneration function in the treatment of myocardial infarction. *Materials Today Bio*, 100978. <https://doi.org/10.1016/j.mtbio.2024.100978>
- Xu, S., Zhang, H., Qian, Z., & Yuan, W. (2024). pH-Responsive injectable self-healing hydrogels loading Au nanoparticles-decorated bimetallic organic frameworks for synergistic sonodynamic-chemodynamic-starvation-chemo therapy of cancer. *Journal of Colloid and Interface Science*, 675, 746–760. <https://doi.org/10.1016/j.jcis.2024.07.039>
- Xu, X., Ho, W., Zhang, X., Bertrand, N., & Farokhzad, O. (2015). Cancer nanomedicine: from targeted delivery to combination therapy. *Trends in Molecular Medicine*, 21(4), 223–232. <https://doi.org/10.1016/j.molmed.2015.01.001>
- Xuan, L., Ju, Z., Skonieczna, M., Zhou, P., & Huang, R. (2023). Nanoparticles-induced potential toxicity on human health: applications, toxicity mechanisms, and evaluation models. *MedComm*, 4(4). <https://doi.org/10.1002/mco2.327>
- Yan, S., Na, J., Liu, X., & Wu, P. (2024). Different targeting ligands-mediated drug delivery systems for tumor therapy. *Pharmaceutics*, 16(2). <https://doi.org/10.3390/PHARMACEUTICS16020248>
- Yanar, F., Carugo, D., & Zhang, X. (2023). Hybrid nanoplatforms comprising organic nanocompartments encapsulating inorganic nanoparticles for enhanced drug delivery and bioimaging applications. *Molecules*, 28(15), 5694. <https://doi.org/10.3390/MOLECULES28155694>
- Yang, E. C., Divine, R., Miranda, M. C., Borst, A. J., Sheffler, W., Zhang, J. Z. et al. (2024). Computational design of non-porous pH-responsive antibody nanoparticles. *Nature Structural & Molecular Biology*, 31(9), 1404–1412. <https://doi.org/10.1038/s41594-024-01288-5>
- Yang, J., & Luo, G. F. (2023). Peptide-based vectors for gene delivery. *Chemistry*, 5(3), 1696–1718. <https://doi.org/10.3390/CHEMISTRY5030116>
- Ye, Q., & Sun, J. (2023). Nanorobots for drug delivery, surgery, and biosensing. In Ki-Taek Lim and Kamel A. Abd-Elsalam (Eds.), *Nanorobotics and Nanodiagnostics in Integrative Biology and Biomedicine*, 15–34. Springer. https://doi.org/10.1007/978-3-031-16084-4_2
- Yetisgin, A. A., Cetinel, S., Zuvin, M., Kosar, A., & Kutlu, O. (2020). Therapeutic nanoparticles and their targeted delivery applications. *Molecules (Basel, Switzerland)*, 25(9). <https://doi.org/10.3390/MOLECULES25092193>

- Yu, M., Wu, J., Shi, J., & Farokhzad, O. C. (2015). Nanotechnology for protein delivery: overview and perspectives. *Journal of Controlled Release*, 240, 24–37. <https://doi.org/10.1016/j.jconrel.2015.10.012>
- Zamboni, W. C., Torchilin, V., Patri, A. K., Hrkach, J., Stern, S., Lee, R. et al. (2012). Best practices in cancer nanotechnology: perspective from NCI nanotechnology alliance. *Clinical Cancer Research: An Official Journal of the American Association for Cancer Research*, 18(12), 3229–3241. <https://doi.org/10.1158/1078-0432.CCR-11-2938>
- Zhang, P., Sun, F., Liu, S., & Jiang, S. (2016). Anti-PEG antibodies in the clinic: current issues and beyond PEGylation. *Journal of Controlled Release*, 244, 184–193. <https://doi.org/10.1016/j.jconrel.2016.06.040>
- Zhang, Q., Dehaini, D., Zhang, Y., Zhou, J., Chen, X., Zhang, L. et al. (2018). Neutrophil membrane-coated nanoparticles inhibit synovial inflammation and alleviate joint damage in inflammatory arthritis. *Nature Nanotechnology*, 13(12), 1182–1190. <https://doi.org/10.1038/s41565-018-0254-4>
- Zhang, S., Gao, H., & Bao, G. (2015). Physical principles of nanoparticle cellular endocytosis. *ACS Nano*, 9(9), 8655–8671. <https://doi.org/10.1021/acs.nano.5b03184>
- Zhang, S., Zhao, B., Jiang, H., Wang, B., & Ma, B. (2007). Cationic lipids and polymers mediated vectors for delivery of siRNA. *Journal of Controlled Release: Official Journal of the Controlled Release Society*, 123(1), 1–10. <https://doi.org/10.1016/J.JCONREL.2007.07.016>
- Zhang, X. D., Wu, D., Shen, X., Liu, P. X., Fan, F. Y., & Fan, S. J. (2012). In vivo renal clearance, biodistribution, toxicity of gold nanoclusters. *Biomaterials*, 33(18), 4628–4638. <https://doi.org/10.1016/J.BIOMATERIALS.2012.03.020>
- Zhao, L., Weir, M. D., & Xu, H. H. (2010). An injectable calcium phosphate-alginate hydrogel-umbilical cord mesenchymal stem cell paste for bone tissue engineering. *Biomaterials*, 31(25), 6502–6510. <https://doi.org/10.1016/j.biomaterials.2010.05.017>
- Zhao, Z., Ukidve, A., Kim, J., & Mitragotri, S. (2020). Targeting strategies for tissue-specific drug delivery. *Cell*, 181(1), 151–167. <https://doi.org/10.1016/J.CELL.2020.02.001>